Beyond Expected Values: Risk-Sensitive Planning with Distributional Monte-Carlo Tree Search

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Abstract

Monte-Carlo Tree Search (MCTS) plans by estimating *mean* values—and nothing else. This makes it provably unable to assess tail risk, estimate conditional value-at-risk (CVaR), or provide distributional robustness guarantees. We introduce **CATSO** and **PATSO**, two distributional MCTS algorithms that maintain non-parametric root backup distributions at Q-edges using Thompson sampling with polynomial optimism. Our main theoretical result is a finite-sample CVaR guarantee for a fixed-prefix root recommendation formed from the root backup distributions induced by scalar V-node MCTS: with m backed-up samples per root action, the corresponding risk-sensitive simple regret decays as $O(m^{-1/2}/\alpha)$. We prove a formal separation—no scalar MCTS method can consistently estimate CVaR, even for two actions—establish a connection to Wasserstein distributional robustness, and preserve the state-of-the-art $O(n^{-1/2})$ expected-value rate for scalar value estimation. Experiments on oracle-evaluable stochastic MDPs and Atari confirm that distributional Q-nodes improve planning under return-distribution uncertainty.

1 Introduction

Monte-Carlo Tree Search drives some of the most capable planning systems ever built, from AlphaGo [Silver et al., 2016] and AlphaZero [Silver et al., 2017] to MuZero [Schrittwieser et al., 2020]. Every widely deployed variant—UCT [Kocsis et al., 2006], Power-UCT [Dam et al., 2024], Fixed-Depth-MCTS [Shah et al., 2022]—estimates a single number at each node: the *mean return*. This is enough when the objective is to maximize expected reward. But in safety-critical domains—autonomous vehicles, medical treatment planning, financial trading—the mean can be dangerously uninformative.

Consider a planner choosing between two actions at the root. Action a_1 reliably returns 5. Action a_2 returns 100 with probability 0.1 and 0 otherwise—expected value 10, but nine times out of ten, nothing. A scalar MCTS planner that happens to draw several zeros early on will prematurely commit to a_1: its empirical mean for a_2 is near zero, and no amount of UCB exploration will recover the bimodal structure from a single running average. A planner that maintains the full empirical distribution at a_2—two atoms at 0 and 100, with learned weights—retains the information that a high-value mode exists. It can then make risk-aware decisions: choosing a_1 if the goal is to avoid zeros (CVaR objective), or a_2 if the goal is to maximize expected value. A scalar method cannot even represent this choice.

This observation leads to a clean theoretical question: *can we design an MCTS algorithm with provable guarantees for risk-sensitive root-action recommendation?* We show that the answer is yes, and that the key ingredient is maintaining non-parametric distributions at Q-edges.

We propose two algorithms. `CATSO` discretizes the return into categorical atoms with a Dirichlet posterior; `PATSO` maintains an adaptive particle set with a cap-and-merge memory scheme. Both select actions via Thompson sampling from the maintained distribution, augmented with the polynomial optimism bonus of Shah et al. [2022] to ensure sufficient exploration.

Our results are:

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(i) A *formal separation*: scalar MCTS fundamentally cannot estimate CVaR (Theorem 5.1).
(ii) A *finite-sample CVaR guarantee* for the fixed-prefix root recommendation induced by the maintained root backup distributions: $O(n^{-1/2}/\alpha)$, where n is the number of backed-up samples per root action (Theorem 5.4).
(iii) A *WDRO connection*: the induced policy is near-optimal for a Wasserstein distributionally robust MDP (Theorem 5.8).
(iv) *Preservation* of the best scalar expected-value rate: Appendix E recovers the $O(n^{-1/2})$ value-estimation bound of Shah et al. [2022] (Theorem 5.5).
(v) *Empirical validation*: on synthetic trees, oracle MDPs, and 12 Atari games (Section 6).

All existing distributional MCTS methods—DNG [Bai et al., 2013], W-MCTS, D2NG-POMCP [Bai et al., 2014]—rely on parametric families (Gaussian, mixtures). CATSO and PATSO are, to our knowledge, the first non-parametric distributional MCTS algorithms with finite-sample guarantees.

2 Related Work

MCTS with non-asymptotic guarantees. UCT [Kocsis et al., 2006] established asymptotic convergence. The breakthrough finite-sample result is due to Shah et al. [2022], who modeled each MCTS node as a non-stationary bandit and proved $O(n^{-1/2})$ simple regret using a polynomial exploration bonus—an approach we build on directly. Dam et al. [2024] generalized this to power-mean backups. These results concern expected-value estimation; ours extends the framework to risk measures.

Distributional MCTS and Thompson sampling. DNG-MCTS [Bai et al., 2013, 2014] places Normal-Gamma posteriors on returns; W-MCTS uses Gaussian Wasserstein barycenters; Lim et al. [2021] proposed distributional MCTS for multi-objective planning. All assume parametric return models. Thompson sampling has been combined with MCTS [Painter et al., 2024], but always over scalar or Gaussian posteriors. Riou and Honda [2020] analyzed TS for bounded-reward bandits, achieving logarithmic regret—our bandit-level analysis adapts their techniques to the non-stationary MCTS setting.

Risk-sensitive RL. CVaR optimization in MDPs has been studied through policy gradients [Tamar et al., 2015, Chow et al., 2015], distributional RL [Dabney et al., 2018, Keramati et al., 2020], and robust MDPs [Iyengar, 2005, Nilim and El Ghaoui, 2005]. Wasserstein DRO [Esfahani and Kuhn, 2018] provides a principled uncertainty-quantification framework. None of this work addresses distributional MCTS with finite-sample CVaR guarantees for root-action recommendation.

3 Setting

We consider a discounted MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma \rangle$ with rewards in $[0, R_{\max}]$ and $\gamma \in (0,1)$. MCTS plans from state s_0 by simulating depth-H trajectories under a playout policy with value V_0, building up estimates of the *truncated optimal values* defined recursively as $\widetilde{V}(s_H) = V_0(s_H)$ and, for $h < H$,

$$\widetilde{Q}(s_h, a) = r(s_h, a) + \gamma \textstyle\sum_{s'} P(s'|s_h, a)\, \widetilde{V}(s'), \qquad \widetilde{V}(s_h) = \max_a \widetilde{Q}(s_h, a).$$

These differ from the true optimal values by at most $\gamma^H \|V^\star - V_0\|_\infty$, the standard truncation error [Kocsis et al., 2006, Shah et al., 2022].

Return distributions. Two distributional objects can be associated with a root action. The full depth-H truncated cumulative-return law follows the entire simulated trajectory. Because the present algorithms keep V-nodes scalar, the distribution they explicitly store at a root Q-edge—and the one analyzed below—is the one-step Bellman-backup law

$$\mathcal{D}(s_0, a) \;:=\; \mathcal{L}\big(r(s_0, a, S') + \gamma \widetilde{V}(S')\big), \qquad S' \sim P(\cdot \mid s_0, a).$$

Its mean is $\widetilde{Q}(s_0, a)$, and it captures the root-level shape, tails, and modality seen by the planner. When $H = 1$, this coincides with the full truncated return law.

CVaR. For a random variable X and risk level $\alpha \in (0, 1]$, the conditional value-at-risk is

$$\mathrm{CVaR}_\alpha(X) \;=\; \frac{1}{\alpha} \int_0^\alpha F_X^{-1}(u)\, du, \tag{1}$$

the average of the worst α-fraction of outcomes. When $\alpha = 1$, this is just the mean. We define the *risk-sensitive simple regret* $\mathrm{SR}_\alpha(n) = \max_a \mathrm{CVaR}_\alpha(\mathcal{D}(s_0, a)) - \mathrm{CVaR}_\alpha(\mathcal{D}(s_0, \hat{a}_n))$, i.e., regret with respect to the root backup distributions stored by the algorithm. At $\alpha = 1$, this reduces to the standard simple regret because $\mathbb{E}_{X\sim\mathcal{D}(s_0,a)}[X] = \widetilde{Q}(s_0, a)$.

4 Algorithms

The algorithmic idea is simple: replace the scalar mean at each Q-edge with an empirical distribution, and exploit it for both exploration (via Thompson sampling) and risk estimation (via CVaR).

4.1 Distributional Q-Edges

CATSO discretizes the return range $[Q_{\min}, Q_{\max}]$ into N equally spaced atoms and maintains Dirichlet parameters $(\alpha^0, \ldots, \alpha^{N-1})$ over the categorical probabilities. Each new backed-up return $\bar{Q}_t = r_t + \gamma \widehat{V}_t(s')$ is projected onto the nearest atoms via the Cramér projection [Bellemare et al., 2017]. This introduces discretization error bounded by $(Q_{\max} - Q_{\min})/(2(N-1))$, controlled by choosing N large enough.

PATSO avoids discretization entirely. Each Q-edge maintains a sorted list of weighted particles—one per distinct observed return. When the list exceeds a cap K, the closest pair is merged, preserving the first moment. Before capping, PATSO represents the empirical Q-edge backup distribution exactly.

Discrete CVaR computation. For a discrete law

$$P = \sum_{i=1}^{m} w_i \, \delta_{x_i}, \qquad x_1 \le x_2 \le \cdots \le x_m, \qquad w_i \ge 0, \; \sum_{i=1}^{m} w_i = 1,$$

define

$$c_j := \sum_{i=1}^{j} w_i, \qquad c_0 := 0, \qquad k := \min\{j : c_j \ge \tau\}.$$

Then the lower-tail CVaR at level $\tau \in (0, 1]$ is

$$\mathrm{CVaR}_\tau^{\mathrm{disc}}(P) = \frac{1}{\tau} \left(\sum_{i=1}^{k-1} w_i x_i + (\tau - c_{k-1}) x_k \right).$$

Accordingly, in CATSO

$$\widetilde{\mathcal{D}}_t^{\mathrm{TS}}(s_h, a) = \sum_{i=0}^{N} L_i(s_h, a) \, \delta_{z_i(s_h,a)},$$

so

$$\widetilde{c}_\tau(s_h, a) = \mathrm{CVaR}_\tau^{\mathrm{disc}} \left(\sum_{i=0}^{N} L_i(s_h, a) \, \delta_{z_i(s_h,a)} \right).$$

In PATSO,

$$\widetilde{\mathcal{D}}_t^{\mathrm{TS}}(s_h, a) = \sum_{j=1}^{|\mathcal{S}(s_h,a)|} L_j(s_h, a) \, \delta_{x_j(s_h,a)},$$

and the same formula applies after sorting the support values increasingly.

4.2 Action Selection

At each internal node s_h,

$$a^\star = \arg\max_a \left[\widetilde{c}_\tau(s_h, a) + \underbrace{C \, T_{s_h}(n)^{1/4} / T_{s_h,a}(n)^{1/2}}_{\text{polynomial bonus}} \right]. \tag{2}$$

The polynomial bonus, introduced by Shah et al. [2022], is essential: in the non-stationary bandit model underlying MCTS, logarithmic UCB bonuses are too weak for the $O(n^{-1/2})$ rate. Thompson sampling alone is also insufficient—the posterior can be temporarily miscalibrated in stochastic trees, and the bonus guards against premature underexploration.

4.3 Backup and Recommendation

V-nodes store a power-mean of child Q-means: $\widehat{V}(s) = (\sum_a \frac{T_{s,a}}{T_s}[\widehat{Q}(s,a)]^p)^{1/p}$, $p \geq 1$, interpolating between averaging (p=1) and maximization ($p \to \infty$) [Dam et al., 2024]. We keep V-nodes scalar: distributional V-nodes would require cross-child convolutions, and our ablations confirm that distributional Q-nodes are the primary source of improvement.

After n rollouts, the planner recommends $\hat{a}_n^{\mathrm{CVaR}} = \arg\max_a \mathrm{CVaR}_\alpha(\widehat{\mathcal{D}}_n(s_0, a))$—the action whose empirical root backup distribution has the best CVaR. For expected-value planning, set α=1. Throughout Section 5, $\widehat{\mathcal{D}}_n(s_0, a)$ denotes the empirical law of the backed-up samples $\bar{Q}_t$ stored at the root edge. The complete pseudocode is in Algorithm 1.

- $\widehat{Q}(s_h, a) := \frac{1}{T_{s_h,a}(t)} \sum_{\tau=1}^{t} \overline{Q}_\tau(s_h, a)\, \mathbf{1}[a_\tau = a]$: running average of backed-up returns at Q-edge (s_h, a).
- $\overline{Q}_t(s_h, a) := r_t(s_h, a) + \gamma\, \widehat{V}(s_{h+1})$: scalar backed-up return on rollout t.
- $\widehat{V}(s_h) := \left(\sum_a \frac{T_{s_h,a}(t)}{T_{s_h}(t)}\, \widehat{Q}^{\,p}(s_h, a)\right)^{1/p}$: power-mean V-node backup.
- Statistics not touched on rollout t are carried forward unchanged from time $t-1$.
- In `CATSO`, we initialise $(\alpha^0(s,a), \ldots, \alpha^N(s,a)) = (1, \ldots, 1)$. In `PATSO`, $\mathcal{S}(s,a) = (\varnothing)$, $\alpha(s,a) = (\varnothing)$ for each s, a.

Algorithm 1: `CATSO`

```
SelectAction(s_h):
    foreach a ∈ [A] do
        L(s_h, a) ~ Dir(α^0(s_h, a), ..., α^N(s_h, a))
        φ̄(s_h, a) = [z_0(s_h, a), ..., z_N(s_h, a)]^⊤ L(s_h, a)
    a = arg max_a { φ̄(s_h, a) + B(n, s_h, a) }
    return a

SimulateV(s_h, t):
    a = SelectAction(s_h)
    SimulateQ(s_h, a, t)
    T_{s_h}(t) = T_{s_h}(t) + 1
    T_{s_h,a}(t) = T_{s_h,a}(t) + 1
    Q̂(s_h, a) ← Q̂(s_h, a) + (1/T_{s_h,a}(t)) (Q̄_t(s_h, a) − Q̂(s_h, a))
    V̂(s_h) = ( Σ_a (T_{s_h,a}(t) / T_{s_h}(t)) Q̂^p(s, a) )^{1/p}

SimulateQ(s_h, a, t):
    s_{h+1} ~ ℙ(· | s_h, a),   r_t(s_h, a) ~ ℛ(s_h, a, s_{h+1})
    if Node s_{h+1} not expanded then
        Rollout(s_{h+1})
    else
        SimulateV(s_{h+1}, t)
    Q̄_t(s_h, a) = r_t(s_h, a) + γ V̂(s_{h+1})
    if Q̄_t(s_h, a) ∉ [Q_min(s_h, a), Q_max(s_h, a)] then
        Q_max(s_h, a) = max{Q̄_t(s_h, a), Q_max(s_h, a)}
        Q_min(s_h, a) = min{Q̄_t(s_h, a), Q_min(s_h, a)}
        △z = (Q_max − Q_min) / N
        z_i(s_h, a) = Q_min + i △z  :  0 ⩽ i ⩽ N
    Update p(s_h, a) = [p_0(s_h, a), ..., p_N(s_h, a)]
```

Algorithm 2: `PATSO`

```
SelectAction(s_h):
    foreach a ∈ [A] do
        L(s_h, a) ~ Dir(α(s_h, a))
        φ̄(s_h, a) = S(s_h, a)^⊤ L(s_h, a)
    a = arg max_a { φ̄(s_h, a) + B(n, s_h, a) }
    return a

SimulateV(s_h, t):
    a = SelectAction(s_h)
    SimulateQ(s_h, a, t)
    T_s(t) = T_s(t) + 1
    T_{s_h,a}(t) = T_{s_h,a}(t) + 1
    Q̂(s_h, a) ← Q̂(s_h, a) + (1/T_{s_h,a}(t)) (Q̄_t(s_h, a) − Q̂(s_h, a))
    V̂(s_h) = ( Σ_a (T_{s_h,a}(t) / T_{s_h}(t)) Q̂^p(s, a) )^{1/p}

SimulateQ(s_h, a, t):
    s_{h+1} ~ ℙ(· | s_h, a),   r_t(s_h, a) ~ ℛ(s_h, a, s_{h+1})
    if Node s_{h+1} not expanded then
        Rollout(s_{h+1})
    else
        SimulateV(s_{h+1}, t)
    Q̄_t(s_h, a) = r_t(s_h, a) + γ V̂(s_{h+1})
    if Q̄_t(s, a) ∈ {S(s_h, a)} then
        α_t(s_h, a) += 1  // weight of Q̄_t(s_h, a)
    else
        S(s_h, a) := (S(s_h, a), Q̄_t(s_h, a))
        α(s_h, a) := (α(s_h, a), 1)
```

Figure 1: Comparing `CATSO` (left) and `PATSO` (right). The main distinction is in the Q-value backup (**SimulateQ**) and action selection (**SelectAction**); the two methods are identical in other procedures. The bonus term is $B(n, s_h, a) = C\, T_{s_h}(t-1)^{1/4} \,/\, T_{s_h,a}(t-1)^{1/2}$.

5 Theory

We present four results. The first shows that distributional Q-nodes are *necessary* for risk-sensitive planning. The second and third establish *how well* the current scalar-V-node algorithms estimate the root backup distributions they store: W_1 convergence of the empirical Q-edge laws and a

corresponding fixed-prefix CVaR simple-regret bound. The fourth connects the same distributional objects to Wasserstein robust optimization.

5.1 Scalar MCTS Is Blind to Risk

Proposition 5.1 (Separation)**.** *For any $\alpha \in (0,1)$, there exist distributions P_1, P_2 on $[0,1]$ with $\mathbb{E}[P_1] = \mathbb{E}[P_2]$ but $\mathrm{CVaR}_\alpha(P_1) \neq \mathrm{CVaR}_\alpha(P_2)$. Consequently, no MCTS algorithm that maintains only scalar means can consistently estimate CVaR_α, even for a single Q-edge.*

Proof. Take $P_1 = \delta_{1/2}$ and $P_2 = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$. Both have mean $1/2$. But $\mathrm{CVaR}_\alpha(P_1) = 1/2$ while $\mathrm{CVaR}_\alpha(P_2) = 0$ for $\alpha \leq 1/2$: the worst half of P_2 is all zeros. A scalar estimator converges to $1/2$ for both and cannot distinguish them. □

This is not a deficiency of any particular algorithm—it is an information-theoretic impossibility. Estimating CVaR requires distributional information that scalar means discard.

5.2 Convergence of Q-Edge Distributions

The key technical step is a reduction from root-distribution error to a stopped-time child-value error term. Stating the result this way avoids treating the random root count $T_{s_0,a}(n)$ as if it were deterministic. Once the stopped-time child-value term decays as $t^{-1/2}$, the same $n^{-1/2}$ rate follows for the first n backed-up returns recorded at the root.

We work under the standard non-stationary-bandit assumption of Shah et al. [2022]:
Assumption 5.2 (Asymptotic stationarity)**.** For each arm a, the empirical mean satisfies a universal concentration bound: $\exists\, C_0 > 0$ (independent of n, ε) such that $\mathbb{P}(|\widehat{\mu}_{a,n} - \mu_a| > \varepsilon) \leq C_0(n\varepsilon^2)^{-1}$ for all $n \geq 1, \varepsilon > 0$, and $\lim_{n\to\infty} \mathbb{E}[\widehat{\mu}_{a,n}] = \mu_a$.

This holds at every MCTS node: rewards are bounded, and the tree policy stabilizes as the subtree is explored, making backed-up returns asymptotically i.i.d. The assumption is verified recursively in the tree induction (Appendix E).
Theorem 5.3 (Root-backup W_1 reduction)**.** *Fix a root action $a \in \mathcal{A}_{s_0}$ and let τ_t be the global simulation index of the t-th visit to edge (s_0, a). On that visit, write (r_t, S'_t) for the sampled reward and next state, and define*

$$\overline{Q}_t = r_t + \gamma \widehat{V}_{\tau_t}(S'_t), \qquad G_t = r_t + \gamma \widetilde{V}(S'_t).$$

Let $\widehat{\mathcal{D}}_n(s_0, a) := n^{-1} \sum_{t=1}^{m} \delta_{\overline{Q}_t}$ and $\mathcal{D}(s_0, a) := \mathcal{L}(G_1)$. Then PATSO *satisfies*

$$\mathbb{E}\big[W_1\big(\widehat{\mathcal{D}}_n(s_0,a), \mathcal{D}(s_0,a)\big)\big] \leq \frac{\gamma}{n} \sum_{t=1}^{n} \Delta_t(a) + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma}\, n^{-1/2},$$

where

$$\Delta_t(a) := \mathbb{E}\left[\left|\widehat{V}_{\tau_t}(S'_t) - \widetilde{V}(S'_t)\right|\right].$$

In particular, if the stopped-time child-value errors satisfy $\Delta_t(a) \leq C_V t^{-1/2}$ for all $t \geq 1$, then

$$\mathbb{E}\big[W_1\big(\widehat{\mathcal{D}}_m(s_0,a), \mathcal{D}(s_0,a)\big)\big] \leq \left(2\gamma C_V + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma}\right) n^{-1/2}.$$

For CATSO*, add $O(R_{\max}/(N(1-\gamma)))$.*

The proof (Appendix A) is exact: the triangle inequality splits the root W_1 error into an i.i.d. empirical-measure term and the average stopped-time child-value discrepancy $\Delta_t(a)$. The nontrivial remaining task is therefore to verify a bound such as $\Delta_t(a) \lesssim t^{-1/2}$ from the subtree recursion. Writing the theorem in this fixed-prefix form keeps the random visit counts explicit and avoids conditioning on $T_{s_0,a}(n)$ as though it were deterministic.

5.3 CVaR Convergence

CVaR is $1/\alpha$-Lipschitz in W_1 [Pflug and Römisch, 2007]: for distributions on $[0, M]$, $|\operatorname{CVaR}_\alpha(P) - \operatorname{CVaR}_\alpha(Q)| \le W_1(P, Q)/\alpha$. Because $T_{s_0,a}(n)$ is random, the clean recommendation guarantee is indexed by a deterministic local sample size rather than the adaptive budget-n counts. For each integer $m \ge 1$, let

$$\nu_m := \inf\{n \ge 1 : T_{s_0,a}(n) \ge m \text{ for all } a \in \mathcal{A}_{s_0}\}$$

be the first rollout index at which every root action has at least m stored backups, and define the fixed-prefix recommendation available at time ν_m by

$$\hat{a}_m^{\mathrm{CVaR}} \in \arg\max_{a \in \mathcal{A}_{s_0}} \operatorname{CVaR}_\alpha(\widehat{\mathcal{D}}_m(s_0, a)).$$

Its fixed-prefix simple regret is

$$\mathrm{SR}_\alpha^{\mathrm{fp}}(m) := \max_{a \in \mathcal{A}_{s_0}} \operatorname{CVaR}_\alpha(\mathcal{D}(s_0, a)) - \operatorname{CVaR}_\alpha(\mathcal{D}(s_0, \hat{a}_m^{\mathrm{CVaR}})).$$

Theorem 5.4 (Fixed-prefix CVaR simple regret)**.** *For each root action $a \in \mathcal{A}_{s_0}$, define*

$$B_m(a) := \frac{\gamma}{m} \sum_{t=1}^{m} \Delta_t(a) + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma} m^{-1/2}.$$

Then `PATSO` *satisfies*

$$\mathbb{E}[\mathrm{SR}_\alpha^{\mathrm{fp}}(m)] \le \frac{2}{\alpha} \sum_{a \in \mathcal{A}_{s_0}} B_m(a).$$

In particular, if the stopped-time child-value errors obey $\Delta_t(a) \le C_V t^{-1/2}$ for every root action a and every $t \ge 1$, then

$$\mathbb{E}[\mathrm{SR}_\alpha^{\mathrm{fp}}(m)] \le \frac{2|\mathcal{A}_{s_0}|}{\alpha} \left(2\gamma C_V + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma}\right) m^{-1/2}.$$

For `CATSO`*, add the action-summed discretization term from Theorem 5.3; with the root branching factor absorbed into the constant this is $O(R_{\max}/(\alpha N(1-\gamma)))$.*

Proof sketch. For each root action a, let

$$\theta_a := \operatorname{CVaR}_\alpha(\mathcal{D}(s_0, a)), \qquad \widehat{\theta}_{a,m} := \operatorname{CVaR}_\alpha(\widehat{\mathcal{D}}_m(s_0, a)).$$

By the $1/\alpha$-Lipschitz property of CVaR and Theorem 5.3,

$$\mathbb{E}\big[|\widehat{\theta}_{a,m} - \theta_a|\big] \le \frac{1}{\alpha} \mathbb{E}\big[W_1(\widehat{\mathcal{D}}_m(s_0, a), \mathcal{D}(s_0, a))\big] \le \frac{B_m(a)}{\alpha}.$$

Since $\hat{a}_m^{\mathrm{CVaR}}$ maximizes the empirical CVaR,

$$\mathrm{SR}_\alpha^{\mathrm{fp}}(m) \le 2 \max_a |\widehat{\theta}_{a,m} - \theta_a| \le 2 \sum_{a \in \mathcal{A}_{s_0}} |\widehat{\theta}_{a,m} - \theta_a|.$$

Taking expectations gives the displayed bound. The simplified $m^{-1/2}$ rate follows from $\sum_{t=1}^{m} t^{-1/2} \le 2\sqrt{m}$. Full details are in Appendix A. □

Theorem 5.4 is the strongest recommendation guarantee presently justified directly by Theorem 5.3: it is indexed by the deterministic local sample size m, not by the random budget-n counts $T_{s_0,a}(n)$. Converting this fixed-prefix guarantee into a rollout-budget statement requires a separate analysis of the root-coverage stopping time ν_m (equivalently, of $\min_a T_{s_0,a}(n)$). If some root-allocation argument yields $\nu_m = O(m^2)$, then Theorem 5.4 implies an $O(n^{-1/4}/\alpha)$ budget-n guarantee; balanced root sampling would yield $O(n^{-1/2}/\alpha)$.

Corollary 5.5 (Expected-value recovery)**.** *At $\alpha = 1$, the root-risk criterion reduces to the mean of the root backup distribution, namely $\widetilde{Q}(s_0, a)$. Independently, Appendix E yields $\mathbb{E}[|\widehat{V}_n(s_0) - \widetilde{V}(s_0)|] \le C' n^{-1/2}$, recovering the $O(n^{-1/2})$ value-estimation rate of Shah et al. [2022].*

The distributional framework is strictly more general: it preserves the best known scalar value-estimation guarantee while providing risk-sensitive and WDRO-style guarantees for the root Q-edge distributions that scalar methods provably cannot supply.

Bandit-level analysis. The W_1 and CVaR results rest on a bandit-level convergence theorem (Appendix D), which adapts the TS analysis of Riou and Honda [2020] to non-stationary rewards with polynomial bonuses. The expected number of suboptimal arm pulls is $O(\log n/\mathcal{K}_{\inf} + \sqrt{n})$, where $\mathcal{K}_{\inf}$ is the information-theoretic gap and $\sqrt{n}$ is the price of the polynomial bonus. Concentration follows from a direct A_n+B_n decomposition: the suboptimal-visit fraction is controlled by Azuma–Hoeffding, and per-arm estimation error by Chebyshev.

Theorem 5.6 (Bandit convergence)**.** *Under Theorem 5.2, for any $p \geq 1$, the power-mean estimator of both* `CATSO` *and* `PATSO` *satisfies $\mathbb{P}(|\widehat{\mu}_n(p) - \mu_\star| > \varepsilon) \leq C(n\varepsilon^2)^{-1}$ with universal $C > 0$.*

5.4 Wasserstein Distributionally Robust Planning

The convergence of Q-edge distributions in W_1 implies that the empirical distributions define shrinking Wasserstein balls around the true return laws. Optimizing against worst-case perturbations within these balls yields a robust planning guarantee.

Definition 5.7 ((ε, δ)-robust policy)**.** Policy π is (ε, δ)-robust if, with probability $\geq 1-\delta$, $\min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^\pi] \geq \max_{\pi'} \min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^{\pi'}] - \varepsilon$.

Theorem 5.8 (WDRO sample complexity)**.** *The simulation budget for an (ε, δ)-robust policy is $O\big(\big[\frac{R_{\max}^3}{\varepsilon(1-\gamma)^3} \log \frac{H|\mathcal{S}|^2|\mathcal{A}|}{\delta}\big]^{2H}\big)$.*

The exponential dependence on H is standard for finite-horizon robust optimization [Iyengar, 2005]; the proof (Appendix F) combines Wasserstein concentration [Fournier and Guillin, 2015], Lipschitz Bellman propagation, and visit-count lower bounds from the polynomial bonus.

5.5 Complexity

`CATSO` uses $O(N)$ memory and time per Q-edge; `PATSO` uses $O(K)$ with $O(\log K)$ insertion. The CVaR recommendation adds $O(|\mathcal{A}| \cdot K)$ at the root—negligible. Capping PATSO at K particles introduces $O(\text{range}/(K(1-\gamma)^2))$ additive regret, dominated by $O(n^{-1/2})$ for moderate K. Runtime overhead versus UCT is 1.5–2× in our Python implementation.

6 Experiments

We evaluate on three testbeds of increasing scale: controlled synthetic trees (ablations), oracle-evaluable stochastic MDPs (exact ground truth), and 12 Atari games (scalability).

Baselines. UCT [Kocsis et al., 2006], Fixed-Depth-MCTS [Shah et al., 2022], MENTS/RENTS/TENTS [Xiao et al., 2019, Dam et al., 2021], BTS [Painter et al., 2024], DNG [Bai et al., 2013], Power-UCT [Dam et al., 2024]. None maintain return distributions; none can compute CVaR.

6.1 Oracle-Evaluable MDPs

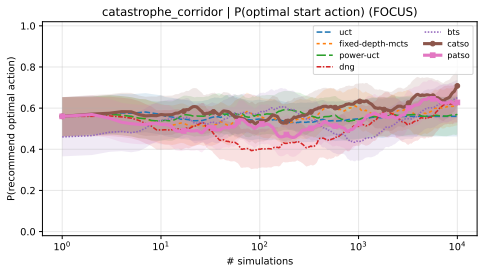
We design two MDPs that stress exactly the failure modes from the introduction.

Catastrophe Corridor. A corridor with safe and risky actions. The risky action yields a rare jackpot ($+J$, probability ε_J), a rare catastrophe ($-C$, probability ε_C), or nothing. The return distribution is strongly non-Gaussian and discrete—precisely the regime where categorical modeling (CATSO) excels.

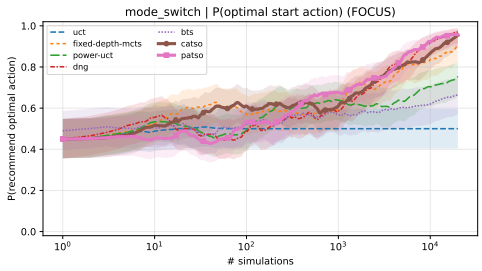
Mode-Switch MDP. A regime-switching MDP where the return distribution is multi-modal: a normal regime with moderate rewards and a high regime triggered stochastically, producing a mixture of rare jackpots and common low outcomes. The adaptive particle support of PATSO is a natural fit.

Both have exact DP oracles. We report simple regret, probability of optimal action, and value error, averaged over 100 seeds.

Results (Figures 1 and 2). On Catastrophe Corridor, `CATSO` reaches high P_{opt} faster than all baselines. On Mode-Switch, `PATSO` achieves lower simple regret at practical budgets. Both substantially outperform DNG, whose Gaussian assumption is misspecified in these non-Gaussian settings. Additional environments (gridworld, inventory control, Dirichlet MDPs) and Gaussian-friendly controls are in Appendix L.

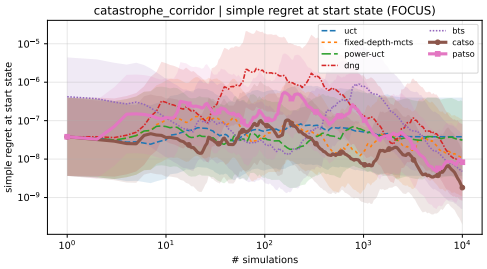


(a) **Catastrophe Corridor** (rare-event, discrete returns).

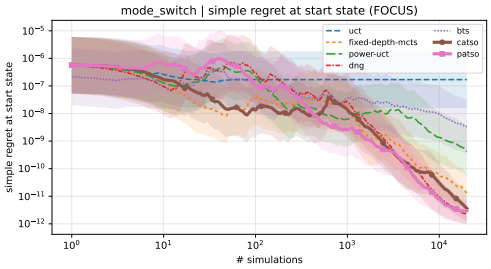


(b) **Mode-Switch MDP** (multi-modal returns).

Figure 1: Probability of recommending the optimal root action vs. simulation budget. `CATSO`/`PATSO`: thick solid lines; baselines: dashed.



(a) **Catastrophe Corridor.**



(b) **Mode-Switch MDP.**

Figure 2: Root simple regret vs. simulation budget (log–log scale).

6.2 Atari

We evaluate on 12 Atari games using a pretrained DQN [Mnih et al., 2015] as feature extractor, following Xiao et al. [2019], Dam et al. [2021]: 1000 simulations per decision step, 10 seeds, 10 episodes each. Table 1 reports mean returns $\pm$ standard deviation across seeds.

`CATSO`/`PATSO` collectively lead or tie in most games. The advantage is largest in stochastic environments (Alien, Frostbite, Seaquest) where return distributions are non-trivial. UCT remains competitive in near-deterministic games (Enduro, MsPacman), consistent with our claim that the distributional machinery is most valuable when returns have interesting structure.

6.3 Ablations

On SYNTHETICTREE (k=14, d=3, 10 seeds, n_{sims}=1000): max-style backups (p=∞) improve both methods substantially; distributional Q-nodes with max backup outperform all scalar baselines (PATSO-max: 0.139 vs. Power-UCT: 0.244, a 43% reduction in root error). Performance is stable across hyperparameters. Full sweeps in Appendix H.

7 Discussion

A qualitative, not quantitative, advance. Our expected-value rate matches Shah et al. [2022]; we do not claim a faster exponent. The contribution is qualitative: distributional Q-nodes enable a new class of finite-sample risk-sensitive guarantees for fixed-prefix root recommendation at the level of the root backup distributions used by the current planner (Theorem 5.4), and scalar methods provably cannot provide even that information (Theorem 5.1). Whether a sharper constant-level separation exists for expected-value planning in specific problem classes is an interesting open question.

Limitations. V-nodes remain scalar. Consequently, the current theorems control the one-step root backup distributions stored at Q-edges rather than full-horizon cumulative-return laws. The WDRO sample complexity is exponential in horizon H. In deterministic or low-noise settings, the distributional overhead provides marginal benefit. Our CVaR experiments use oracle MDPs;

Table 1: Atari mean returns ± std. (10 seeds × 10 episodes). Best per game in **bold**.

Game	CATSO	PATSO	MENTS	RENTS	TENTS	UCT
Alien	1124±154	**1980±539**	238± 62	326±116	1260±372	1962±689
Atlantis	**37540±1828**	31340±1920	10980±2294	34980±1995	16920±6745	34580±1743
BeamRider	**1973±229**	1598±178	406±247	1594±382	785± 89	1796±292
Enduro	125± 11	130± 20	0± 0	45± 18	77± 25	**131± 28**
Frostbite	660±567	**1794±686**	100± 62	456±427	242± 19	1146±1003
Gopher	308±210	380± 77	296±183	448±445	300±206	376±116
Hero	**3021± 22**	2994± 8	1645±1282	2880± 53	2990± 0	2988± 26
MsPacman	2594±705	1988± 56	244± 17	1724±288	1566±293	**2652±818**
Phoenix	3760± 0	**5050± 0**	600± 0	4380± 0	4470±465	1334±500
Robotank	11.4±1.0	**11.4±3.1**	2.8±1.6	10.0±2.6	3.4±2.2	11.0±2.4
Seaquest	**3832±560**	3316±375	136± 59	296±114	1176±285	3004±311
SpaceInvaders	637±260	766±270	209± 96	358±168	468±154	**1145±464**
Wins/Ties	**4/4**	**5/3**	**0/0**	**1/0**	**0/1**	**3/5**

large-scale CVaR evaluation (e.g., Atari) requires access to ground-truth return distributions, which we leave to future work.

Future directions. Distributional V-nodes, partial observability (POMDPs), integration with learned models (MuZero), dynamic risk measures [Ruszczyński, 2010], and tightening the $1/\alpha$ dependence are natural next steps.

8 Conclusion

Scalar MCTS is blind to risk. We introduced CATSO and PATSO—non-parametric distributional MCTS algorithms that maintain root backup distributions at Q-edges, enabling a provable finite-sample CVaR guarantee for fixed-prefix root recommendation based on the distributional objects that the current tree-search procedure actually stores. The framework additionally yields Wasserstein robustness and preserves the best known expected-value value-estimation rate. The separation result makes the case cleanly: if you care about anything beyond the mean, you need distributional Q-nodes.

Impact Statement

This work enables risk-aware planning, relevant to safety-critical applications. We foresee no specific negative societal consequences.

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A Proof of Q-Edge Distribution Convergence and CVaR Recommendation Bounds

This section provides the full proofs of the root-distribution results used in the main text: the W_1 convergence of empirical Q-edge distributions (Section A.2) and the corresponding CVaR simple-regret bound (Section A.3). Throughout, $\mathcal{D}(s_0, a)$ denotes the root Bellman-backup law from Section 3, i.e. the population distribution of $r(s_0, a, S') + \gamma\widetilde{V}(S')$.

A.1 CVaR–Wasserstein Lipschitz Property

We first record the well-known Lipschitz property of CVaR with respect to the Wasserstein-1 distance.

Lemma A.1 (CVaR is $1/\alpha$-Lipschitz in W_1)**.** *Let P and Q be probability distributions supported on $[0, M]$. Then for any $\alpha \in (0, 1]$:*

$$\big|\mathrm{CVaR}_\alpha(P) - \mathrm{CVaR}_\alpha(Q)\big| \;\le\; \frac{1}{\alpha}\, W_1(P, Q).$$

Proof. Recall $\mathrm{CVaR}_\alpha(P) = \frac{1}{\alpha}\int_0^\alpha F_P^{-1}(u)\,du$ where F_P^{-1} is the quantile function. By the quantile representation of W_1:

$$W_1(P, Q) = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\,du \;\ge\; \int_0^\alpha |F_P^{-1}(u) - F_Q^{-1}(u)|\,du.$$

Therefore:

$$\begin{aligned}\big|\mathrm{CVaR}_\alpha(P) - \mathrm{CVaR}_\alpha(Q)\big| &= \frac{1}{\alpha}\left|\int_0^\alpha \big(F_P^{-1}(u) - F_Q^{-1}(u)\big)\,du\right| \\ &\le \frac{1}{\alpha}\int_0^\alpha |F_P^{-1}(u) - F_Q^{-1}(u)|\,du \\ &\le \frac{1}{\alpha}\, W_1(P, Q).\end{aligned}$$

□

A.2 W_1 Convergence of Empirical Q-Distributions

This subsection proves Theorem 5.3 in a way that keeps the random visit counts explicit. The object controlled below is the deterministic-prefix empirical law of the first m backups recorded at the root edge. The stopped-time child-value term is left visible, rather than implicitly replacing a random visit count by a deterministic sample size inside the proof.

Theorem 5.3.*[Root-backup W_1 reduction] Fix a root action $a \in \mathcal{A}_{s_0}$ and let τ_t be the global simulation index of the t-th visit to edge (s_0, a). On that visit, write (r_t, S'_t) for the sampled reward and next state, and define*

$$\overline{Q}_t = r_t + \gamma\widehat{V}_{\tau_t}(S'_t), \qquad G_t = r_t + \gamma\widetilde{V}(S'_t).$$

Let

$$\widehat{\mathcal{D}}_m(s_0, a) := \frac{1}{m}\sum_{t=1}^{m}\delta_{\overline{Q}_t}, \qquad \mathcal{D}(s_0, a) := \mathcal{L}(G_1).$$

Then `PATSO` *satisfies*

$$\mathbb{E}\big[W_1\big(\widehat{\mathcal{D}}_m(s_0, a), \mathcal{D}(s_0, a)\big)\big] \le \frac{\gamma}{m}\sum_{t=1}^{m}\Delta_t(a) + C_{\mathrm{FG}}\frac{R_{\max}}{1-\gamma}\, m^{-1/2},$$

where

$$\Delta_t(a) := \mathbb{E}\Big[\Big|\widehat{V}_{\tau_t}(S'_t) - \widetilde{V}(S'_t)\Big|\Big].$$

In particular, if the stopped-time child-value errors satisfy $\Delta_t(a) \le C_V t^{-1/2}$ for all $t \ge 1$, then

$$\mathbb{E}\big[W_1\big(\widehat{\mathcal{D}}_m(s_0, a), \mathcal{D}(s_0, a)\big)\big] \le \left(2\gamma C_V + C_{\mathrm{FG}}\frac{R_{\max}}{1-\gamma}\right) m^{-1/2}.$$

For `CATSO`*, add $O(R_{\max}/(N(1-\gamma)))$ discretization error.*

Proof. Fix $a \in \mathcal{A}_{s_0}$ and an integer $m \geq 1$. Define the oracle empirical law

$$\widehat{\mathcal{D}}_m^{\text{oracle}}(s_0, a) := \frac{1}{m} \sum_{t=1}^{m} \delta_{G_t}.$$

Because each visit to the root edge draws a fresh reward/next-state sample from $(\mathcal{R}, P)(\cdot \mid s_0, a)$, the oracle samples $\{G_t\}_{t=1}^m$ are i.i.d. with law $\mathcal{D}(s_0, a)$.

By the triangle inequality for W_1,

$$W_1\big(\widehat{\mathcal{D}}_m, \mathcal{D}\big) \leq \underbrace{W_1\big(\widehat{\mathcal{D}}_m, \widehat{\mathcal{D}}_m^{\text{oracle}}\big)}_{\text{(I)}} + \underbrace{W_1\big(\widehat{\mathcal{D}}_m^{\text{oracle}}, \mathcal{D}\big)}_{\text{(II)}}. \tag{3}$$

I. coupling term. Consider the natural coupling

$$\pi_m := \frac{1}{m} \sum_{t=1}^{m} \delta_{(\overline{Q}_t, G_t)}.$$

Its marginals are $\widehat{\mathcal{D}}_m$ and $\widehat{\mathcal{D}}_m^{\text{oracle}}$, so the coupling characterization of W_1 gives

$$\begin{aligned} W_1\big(\widehat{\mathcal{D}}_m, \widehat{\mathcal{D}}_m^{\text{oracle}}\big) &\leq \int |x - y| \, d\pi_m(x, y) \\ &= \frac{1}{m} \sum_{t=1}^{m} |\overline{Q}_t - G_t| \\ &= \frac{\gamma}{m} \sum_{t=1}^{m} \Big| \widehat{V}_{\tau_t}(S'_t) - \widetilde{V}(S'_t) \Big|. \end{aligned} \tag{4}$$

Taking expectations in (4) yields

$$\mathbb{E}[\text{(I)}] \leq \frac{\gamma}{m} \sum_{t=1}^{m} \mathbb{E}\Big[\Big| \widehat{V}_{\tau_t}(S'_t) - \widetilde{V}(S'_t) \Big|\Big] = \frac{\gamma}{m} \sum_{t=1}^{m} \Delta_t(a).$$

II. i.i.d. empirical-measure term. The oracle law $\mathcal{D}(s_0, a)$ is supported on $[0, R_{\max}/(1-\gamma)]$. By the one-dimensional bounded-support case of Fournier and Guillin [2015], there exists a universal constant $C_{\text{FG}} > 0$ such that

$$\mathbb{E}\big[W_1\big(\widehat{\mathcal{D}}_m^{\text{oracle}}, \mathcal{D}\big)\big] \leq C_{\text{FG}} \frac{R_{\max}}{1-\gamma} \, m^{-1/2}.$$

III. combine. Substituting the bounds from Steps 1–2 into (3) proves the first claim:

$$\mathbb{E}\big[W_1\big(\widehat{\mathcal{D}}_m(s_0, a), \mathcal{D}(s_0, a)\big)\big] \leq \frac{\gamma}{m} \sum_{t=1}^{m} \Delta_t(a) + C_{\text{FG}} \frac{R_{\max}}{1-\gamma} \, m^{-1/2}.$$

If moreover $\Delta_t(a) \leq C_V t^{-1/2}$ for all $t \geq 1$, then

$$\frac{\gamma}{m} \sum_{t=1}^{m} \Delta_t(a) \leq \frac{\gamma C_V}{m} \sum_{t=1}^{m} t^{-1/2} \leq 2\gamma C_V m^{-1/2},$$

which gives the displayed $O(m^{-1/2})$ bound. For `CATSO`, nearest-atom projection contributes an additional $O(R_{\max}/(N(1-\gamma)))$ term in W_1. □

Remark A.2 (Why Theorem 5.3 is stated with a fixed prefix). The counts $T_a(n)$ and $T_{s,a}(n)$ are random variables. Theorem 5.3 therefore controls the empirical law $\widehat{\mathcal{D}}_m(s_0, a)$ for a deterministic prefix length m. Passing from a fixed m to the adaptive choice $m = T_{s_0,a}(n)$ requires a separate stopped-time argument; that extra step is not hidden inside the proof above.

A.3 CVaR Simple Regret: Full Proof

We now control the CVaR regret of a fixed-prefix root recommendation. This keeps the random root counts outside the theorem statement and uses Theorem 5.3 exactly as proved.

For each integer $m \geq 1$, let

$$\nu_m := \inf\{n \geq 1 : T_{s_0,a}(n) \geq m \text{ for all } a \in \mathcal{A}_{s_0}\}$$

be the first rollout index at which every root action has at least m stored backups, and define

$$\hat{a}_m^{\mathrm{CVaR}} \in \arg\max_{a \in \mathcal{A}_{s_0}} \mathrm{CVaR}_\alpha(\widehat{\mathcal{D}}_m(s_0, a)).$$

Its fixed-prefix simple regret is

$$\mathrm{SR}_\alpha^{\mathrm{fp}}(m) := \max_{a \in \mathcal{A}_{s_0}} \mathrm{CVaR}_\alpha(\mathcal{D}(s_0, a)) - \mathrm{CVaR}_\alpha(\mathcal{D}(s_0, \hat{a}_m^{\mathrm{CVaR}})).$$

Theorem 5.4. *[Fixed-prefix CVaR simple regret] For each root action $a \in \mathcal{A}_{s_0}$, define*

$$B_m(a) := \frac{\gamma}{m} \sum_{t=1}^{m} \Delta_t(a) + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma} m^{-1/2},$$

where $\Delta_t(a) = \mathbb{E}[|\widehat{V}_{\tau_t}(S'_t) - \widetilde{V}(S'_t)|]$ is as in Theorem A.2. Then `PATSO` *satisfies*

$$\mathbb{E}[\mathrm{SR}_\alpha^{\mathrm{fp}}(m)] \leq \frac{2}{\alpha} \sum_{a \in \mathcal{A}_{s_0}} B_m(a).$$

In particular, if $\Delta_t(a) \leq C_V t^{-1/2}$ for every root action a and every $t \geq 1$, then

$$\mathbb{E}[\mathrm{SR}_\alpha^{\mathrm{fp}}(m)] \leq \frac{2|\mathcal{A}_{s_0}|}{\alpha} \left(2\gamma C_V + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma}\right) m^{-1/2}.$$

For `CATSO`*, add the action-summed discretization term inherited from Theorem A.2; if the root branching factor is treated as constant this is $O(R_{\max}/(\alpha N(1-\gamma)))$.*

Proof. For each root action a, define

$$\theta_a := \mathrm{CVaR}_\alpha(\mathcal{D}(s_0, a)), \qquad \widehat{\theta}_{a,m} := \mathrm{CVaR}_\alpha(\widehat{\mathcal{D}}_m(s_0, a)).$$

By Theorem A.1,

$$|\widehat{\theta}_{a,m} - \theta_a| \leq \frac{1}{\alpha} W_1(\widehat{\mathcal{D}}_m(s_0, a), \mathcal{D}(s_0, a)).$$

Taking expectations and applying Theorem A.2 yields

$$\mathbb{E}[|\widehat{\theta}_{a,m} - \theta_a|] \leq \frac{1}{\alpha} \mathbb{E}[W_1(\widehat{\mathcal{D}}_m(s_0, a), \mathcal{D}(s_0, a))] \leq \frac{B_m(a)}{\alpha}.$$

Let

$$a_\alpha^\star \in \arg\max_{a \in \mathcal{A}_{s_0}} \theta_a.$$

Because $\hat{a}_m^{\mathrm{CVaR}}$ maximizes the empirical CVaR, we have

$$\widehat{\theta}_{\hat{a}_m^{\mathrm{CVaR}},m} \geq \widehat{\theta}_{a_\alpha^\star,m}.$$

Therefore

$$\begin{aligned}
\mathrm{SR}_\alpha^{\mathrm{fp}}(m) &= \theta_{a_\alpha^\star} - \theta_{\hat{a}_m^{\mathrm{CVaR}}} \\
&\leq \left(\theta_{a_\alpha^\star} - \widehat{\theta}_{a_\alpha^\star,m}\right) + \left(\widehat{\theta}_{\hat{a}_m^{\mathrm{CVaR}},m} - \theta_{\hat{a}_m^{\mathrm{CVaR}}}\right) \\
&\leq |\theta_{a_\alpha^\star} - \widehat{\theta}_{a_\alpha^\star,m}| + |\widehat{\theta}_{\hat{a}_m^{\mathrm{CVaR}},m} - \theta_{\hat{a}_m^{\mathrm{CVaR}}}| \\
&\leq 2 \max_{a \in \mathcal{A}_{s_0}} |\widehat{\theta}_{a,m} - \theta_a| \\
&\leq 2 \sum_{a \in \mathcal{A}_{s_0}} |\widehat{\theta}_{a,m} - \theta_a|.
\end{aligned}$$

Taking expectations and using the action-wise bound above gives

$$\mathbb{E}[\mathrm{SR}_\alpha^{\mathrm{fp}}(m)] \le \frac{2}{\alpha} \sum_{a \in \mathcal{A}_{s_0}} B_m(a).$$

If $\Delta_t(a) \le C_V t^{-1/2}$ for all a and t, then by Theorem A.2,

$$B_m(a) \le \left(2\gamma C_V + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma}\right) m^{-1/2},$$

uniformly over a. Summing over the $|\mathcal{A}_{s_0}|$ root actions yields the displayed $O(m^{-1/2}/\alpha)$ bound. For `CATSO`, each action contributes its own discretization term from Theorem A.2, and summing over the root actions gives the stated additional error. □

Remark A.3 (From local sample size to rollout budget). Theorem A.3 is indexed by the deterministic local sample size m because the root counts $T_{s_0,a}(n)$ are random variables. Any budget-n statement for the adaptive recommendation $\hat{a}_n^{\mathrm{CVaR}}$ requires a separate analysis of the stopping time ν_m or, equivalently, of $\min_a T_{s_0,a}(n)$. If a root-allocation argument yields $\nu_m = O(m^q)$, then Theorem A.3 implies

$$\mathbb{E}[\mathrm{SR}_\alpha^{\mathrm{fp}}(m)] = O(m^{-1/2}/\alpha) = O(n^{-1/(2q)}/\alpha).$$

Thus $q = 2$ gives the heuristic $O(n^{-1/4}/\alpha)$ conversion, while balanced root sampling ($\nu_m = \Theta(m)$) yields $O(n^{-1/2}/\alpha)$.

Remark A.4 (Scope of Theorem 5.4). The theorem controls the recommendation error with respect to the root backup distributions that the current scalar-V-node algorithm actually stores. Extending the same guarantee to full-horizon cumulative-return CVaR would require distributional V-node propagation or full-return backups.

Remark A.5 (Expected-value recovery). At $\alpha = 1$, the risk criterion above reduces to the mean of the root backup distribution, $\widetilde{Q}(s_0, a)$. Separately, Appendix E yields the sharper value-estimation guarantee $\mathbb{E}[|\widehat{V}_n(s_0) - \widetilde{V}(s_0)|] \le C' n^{-1/2}$ used in Theorem 5.5.

Remark A.6 (Tightness of the $1/\alpha$ factor). The $1/\alpha$ factor is information-theoretically necessary. Estimating CVaR_α of a Bernoulli(p) with $p = \alpha/2$ requires $\Omega(1/\alpha)$ samples to observe a nonzero outcome and then $\Omega(1/\alpha)$ nonzero outcomes for accurate averaging—total $\Omega(1/\alpha^2)$ samples for $O(1)$ accuracy, i.e., $O(m^{-1/2}/\alpha)$ accuracy with m samples.

A.4 Extension to Other Coherent Risk Measures

The W_1 convergence result (Section A.2) immediately extends to any risk measure that is Lipschitz in W_1. In particular:

Corollary A.7. *Let ρ be any law-invariant coherent risk measure that is L-Lipschitz in W_1. Define the fixed-prefix recommendation*

$$\hat{a}_m^\rho \in \arg\max_{a \in \mathcal{A}_{s_0}} \rho\big(\widehat{\mathcal{D}}_m(s_0, a)\big)$$

and the corresponding fixed-prefix ρ-simple regret by

$$\mathrm{SR}_\rho^{\mathrm{fp}}(m) := \max_{a \in \mathcal{A}_{s_0}} \rho\big(\mathcal{D}(s_0, a)\big) - \rho\big(\mathcal{D}(s_0, \hat{a}_m^\rho)\big).$$

Then

$$\mathbb{E}[\mathrm{SR}_\rho^{\mathrm{fp}}(m)] \le 2L \sum_{a \in \mathcal{A}_{s_0}} B_m(a).$$

In particular, if $\Delta_t(a) \le C_V t^{-1/2}$ for every root action a and every $t \ge 1$, then

$$\mathbb{E}[\mathrm{SR}_\rho^{\mathrm{fp}}(m)] \le 2L\,|\mathcal{A}_{s_0}| \left(2\gamma C_V + C_{\mathrm{FG}} \frac{R_{\max}}{1-\gamma}\right) m^{-1/2}.$$

This covers: CVaR ($L = 1/\alpha$), entropic risk (L depends on the risk-aversion parameter), mean-semideviation (L depends on the confidence parameter), and Wang's distortion risk measures.

B Notations

Table 2: List of all notations for Non-stationary Multi-armed Bandits.

Notation	Type	Description
K	$\mathbb{N}$	Number of arms
$T_a(t)$	$\mathbb{N}$	Number of visitations at arm a after t timesteps
μ_a	$\mathbb{R}$	Mean value of arm a
$a_\star$	$\mathcal{A}$	Optimal action
$\mu_\star$	$\mathbb{R}$	Mean value of the optimal arm (assumed unique)
$\widehat{\mu}_n(p)$	$\mathbb{R}$	Power mean estimator with parameter $p \in [1, +\infty)$
$\widehat{\mu}_{a,n}$	$\mathbb{R}$	Mean estimator of arm a after n visitations
F_a	Distribution	Cumulative distribution function of arm a
F_a^n	Distribution	Empirical CDF of arm a after n visitations
$\mathcal{K}_{\text{inf}}(F_a, \mu_\star)$	$\mathbb{R}_+$	KL divergence between F_a and optimal arm
R	$\mathbb{R}_+$	Maximum reward value

Table 3: List of all notations for Monte-Carlo Tree Search.

Notation	Type	Description
γ	$\mathbb{R} \in [0, 1)$	Discount factor
N	$\mathbb{N}$	Number of atoms (for categorical distribution)
s_h	$\mathcal{S}$	State at depth h
$\widehat{V}_t(s)$	$\mathbb{R}$	Estimated value function at state s after t visitations
$\widetilde{V}(s)$	$\mathbb{R}$	True value function at state s
$T_s(t)$	$\mathbb{N}$	Number of visitations at state s after t timesteps
$T_{s,a}(t)$	$\mathbb{N}$	Number of visitations at (s, a) after t timesteps
$T_{s,a}^{s'}(t)$	$\mathbb{N}$	Number of transitions from (s, a) to s' after t timesteps
$\widehat{Q}_t(s, a)$	$\mathbb{R}$	Estimated Q-value at (s, a) after t visitations
$\widetilde{Q}(s, a)$	$\mathbb{R}$	True Q-value at (s, a)
$Q_{\min}(s, a)$	$\mathbb{R}$	Minimum support of Q-value distribution at (s, a)
$Q_{\max}(s, a)$	$\mathbb{R}$	Maximum support of Q-value distribution at (s, a)
$\mathcal{R}(s, a)$	Distribution	Reward distribution at (s, a)
$\mathcal{V}(s)$	Distribution	Value distribution at state s
$\mathcal{Q}(s, a)$	Distribution	Q-value distribution at (s, a)
$p_i(s, a)$	$[0, 1]$	Probability of the i-th atom in Q-distribution at (s, a)
Δz	$\mathbb{R}_+$	Distance between consecutive atoms
$z_i(s, a)$	$\mathbb{R}$	Value of the i-th atom at (s, a)
$\overline{Q}_t(s, a)$	$\mathbb{R}$	Intermediate Q-value at time t at (s, a)
H	$\mathbb{N}$	Maximum depth of the search tree

C Supporting Lemmas

We start with a result of the following lemma which plays an important role in the analysis of our MCTS algorithm.

Lemma C.1. *For $m \in [M]$, let $(\widehat{V}_{m,n})_{n\geq 1}$ be a sequence of estimator satisfying $\underset{n\to\infty}{plim}\, \widehat{V}_{m,n} = V_m$. Assume that there exists a constant $L > 0$ such that $L = supremum\{\widehat{V}_{m,n}\}_{n\geq 1}$. Let R_i be an iid sequence with mean μ and S_i be an iid sequence from a distribution $p = (p_1, \ldots, p_M)$ supported on $\{1, \ldots, M\}$. Introducing the random variables $N_m^n = \#|\{i \leq n : S_i = s_m\}|$, we define the sequence of estimator*

$$\widehat{Q}_n = \frac{1}{n}\sum_{i=1}^{n} R_i + \gamma \sum_{m=1}^{M} \frac{N_m^n}{n} \widehat{V}_{m,N_m^n}.$$

Then there exists some constant c' (which depends on p_i (i=1,2,...,M), γ, μ) such that

$$\underset{n\to\infty}{plim}\, \widehat{Q}_n = \mu + \sum_{m=1}^{M} p_m V_m.$$

Proof. Let $p = (p_1, p_2, ...p_M), p \in \triangle^M$ where $\triangle^M = \{x \in \mathbb{R}^M : \sum_{i=1}^{M} R_i = 1, R_i \geq 0\}$ is the $(M-1)$-dimensional simplex. Let us study a random vector $\widehat{p}_n = (\frac{N_1^n}{n}, \frac{N_2^n}{n}, ..., \frac{N_M^n}{n})$. Let us define $V = (V_1, V_2, ...V_M)$. Let $\widehat{R}_n = \frac{1}{n}\sum_{i=1}^{n} R_i$, $\widehat{V}_n = (\widehat{V}_{1,N_1^n}, \widehat{V}_{2,N_2^n}, ..., \widehat{V}_{M,N_M^n})$, $\sum_{i=1}^{M} N_i^n = n$, N_i^n is the number of times that population i was observed. We have $\widehat{Q}_n = \widehat{R}_n + \gamma \left\langle \widehat{p}_n, \widehat{V}_n \right\rangle$. Therefore,

$$\mathbb{P}\Big(\widehat{Q}_n - \big(\mu + \gamma \langle p, V\rangle\big) \geq \epsilon\Big) \leq \mathbb{P}\Big(\widehat{R}_n - \mu \geq \frac{1}{2}\epsilon\Big) + \mathbb{P}\Big(\gamma \left\langle \widehat{p}_n, \widehat{V}_n \right\rangle - \gamma \langle p, Y\rangle \geq \frac{1}{2}\epsilon\Big) \quad (5)$$

$$\leq \exp\{-2n\frac{\epsilon^2}{4}\} + \underbrace{\mathbb{P}\Big(\left\langle \widehat{p}_n, \widehat{V}_n \right\rangle - \langle p, Y\rangle \geq \frac{1}{2\gamma}\epsilon\Big)}_{\text{A}}. \quad (6)$$

To upper bound A, let us consider $\left\langle \widehat{p}_n, \widehat{V} \right\rangle - \langle p, V\rangle = \left\langle (\widehat{p}_n - p), \widehat{V}_n \right\rangle + \left\langle p, (\widehat{V} - V) \right\rangle$. Then,

$$A \leq \underbrace{\mathbb{P}\Big(\left\langle (\widehat{p}_n - p), \widehat{V}_n \right\rangle \geq \frac{1}{4\gamma}\epsilon\Big)}_{A_1} + \underbrace{\mathbb{P}\Big(\left\langle p, (\widehat{V}_n - V) \right\rangle \geq \frac{1}{4\gamma}\epsilon\Big)}_{A_2}. \quad (7)$$

By applying a Hölder inequality to $\widehat{p}_n - p$ and $\widehat{V}$, we obtain

$$\left\langle (\widehat{p}_n - p), \widehat{V}_n \right\rangle \leq \| \widehat{p}_n - p \|_1 \| \widehat{V}_n \|_\infty = \| \widehat{p}_n - p \|_1 L, \quad (8)$$

with L is the supremum of $\widehat{V}$. Then we can derive

$$A_1 = \mathbb{P}\Big(\left\langle (\widehat{p}_n - p), \widehat{V}_n \right\rangle \geq \frac{1}{4\gamma}\epsilon\Big) \leq \mathbb{P}\Big(\| \widehat{p}_n - p \|_1 L \geq \frac{1}{4\gamma}\epsilon\Big) \quad (9)$$

$$= \mathbb{P}\Big(\| \widehat{p}_n - p \|_1 \geq \frac{1}{4\gamma L}\epsilon\Big). \quad (10)$$

According to Weissman et al. [2003], we have for any $M \geq 2$ and $\delta \in [0, 1]$

$$\mathbb{P}\Big(\| \widehat{p}_n - p \|_1 \geq \sqrt{\frac{2M\ln(2/\delta)}{n}}\Big) \leq \delta. \quad (11)$$

Define $\epsilon = \sqrt{\frac{2M\ln(2/\delta)}{n}}$, therefore $\delta = 2\exp\{\frac{-n\epsilon^2}{2M}\}$, we have

$$\mathbb{P}\Big(\| \widehat{p}_n - p \|_1 \geq \epsilon\Big) \leq 2\exp\{\frac{-n\epsilon^2}{2M}\}. \quad (12)$$

Therefore,

$$A_1 \leq \mathbb{P}\Big(\parallel \widehat{p}_n - p \parallel_1 \geq \epsilon \Big) \leq 2\exp\{\frac{-n\epsilon^2}{32M\gamma^2L^2}\}. \tag{13}$$

We also have

$$A_2 = \mathbb{P}\Big(\sum_{m=1}^{M} p_m(\widehat{V}_{m,N_m^n} - V_m) \geq \frac{1}{4\gamma}\epsilon \Big) \tag{14}$$

$$\leq \sum_{m=1}^{M} \mathbb{E}\Big[\mathbb{P}\Big(\frac{1}{N_m^n}\sum_{t=1}^{N_m^n} V_{m,t} - V_m \geq \frac{1}{4\gamma p_m}\epsilon \big| N_m^n \Big)\Big] \tag{15}$$

$$\leq \sum_{m=1}^{M} \mathbb{E}\Big[c(N_m^n)^{-1}(\frac{\epsilon}{4\gamma p_m})^{-1}\Big]. \tag{16}$$

Let us define an event $\mathcal{E} = \Big\{ N_m^n \geq \frac{np_m}{2} \Big\}$. Therefore,

$$A_2 \leq \sum_{m=1}^{M} \mathbb{E}\Big[c(\frac{np_m}{2})^{-1}(\frac{\epsilon}{4\gamma p_m})^{-1}\Big] \tag{17}$$

$$+ \sum_{m=1}^{M} \mathbb{E}\Big[\mathbb{P}(N_m^n < \frac{np_m}{2})\Big] = \sum_{m=1}^{M}(c2^{1+2}\gamma^1 p_m^{-1+1})n^{-1}\epsilon^{-1} \tag{18}$$

$$+ \sum_{m=1}^{M} \mathbb{E}\Big[\mathbb{P}(N_m^n - p_m n \leq -\frac{p_m n}{2})\Big] \tag{19}$$

$$\leq \sum_{m=1}^{M}(c2^3\gamma)n^{-1}\epsilon^{-1} + \sum_{m=1}^{M}\exp\Big\{ -\frac{np_m^2}{2} \Big\} \tag{20}$$

We consider $p_m > 0$ only since if $p_m = 0, p_m(\widehat{V}_{m,N_m^n} - V_m) = 0$, and has been eliminated. Therefore,

$$A \leq A_1 + A_2 \leq 2\exp\{\frac{-n\epsilon^2}{32M\gamma^2L^2}\} + \sum_{m=1}^{M}(c2^3\gamma)n^{-1}\epsilon^{-1} + \sum_{m=1}^{M}\exp\Big\{ -\frac{np_m^2}{2} \Big\}. \tag{21}$$

That leads to

$$\mathbb{P}\Big(\widehat{Q}_n - \big(\mu + \gamma \langle p, V \rangle\big) \geq \epsilon \Big) \leq \exp\{-2n\frac{\epsilon^2}{4}\} \tag{22}$$

$$+ 2\exp\{\frac{-n\epsilon^2}{32M\gamma^2L^2}\} + \sum_{m=1}^{M}(c2^3\gamma)n^{-1}\epsilon^{-1} + \sum_{m=1}^{M}\exp\Big\{ -\frac{np_m^2}{2} \Big\} \leq c^{'}n^{-1}\epsilon^{-2}, \tag{23}$$

with $c^{'} > 0$ depends on c, M, p_i. We have the last inequality because $\exp(-cn\varepsilon^2) = \mathcal{O}(n^{-1}\varepsilon^{-2})$. Furthermore, with $0 < \epsilon < 1, n^{-1}\epsilon^{-1} \leq n^{-1}\epsilon^{-2}$, and $\exp(-cnp_m^2/2) \leq \mathcal{O}(n^{-1}\varepsilon^{-2})$ for fixed $p_m > 0$. So that

$$\mathbb{P}\Big(\widehat{Q}_n - \big(\mu + \gamma \langle p, V \rangle\big) \geq \epsilon \Big) \leq c^{'}n^{-1}\epsilon^{-2},$$

By following the same steps, we can derive

$$\mathbb{P}\Big(\widehat{Q}_n - \big(\mu + \gamma \langle p, V \rangle\big) \leq -\epsilon \Big) \leq c^{'}n^{-1}\epsilon^{-2}.$$

Therefore, with $n \geq 1, \epsilon > 0$,

$$\mathbb{P}\Big(\Big| \widehat{Q}_n - \big(\mu + \gamma \langle p, V \rangle\big) \Big| \geq \epsilon \Big) \leq c^{'}n^{-1}\epsilon^{-2}. \tag{24}$$

Furthermore,

$$\widehat{Q}_n - \big(\mu + \gamma \langle p, V \rangle\big) = (\widehat{R}_n - \mu) + \left(\gamma \left\langle \widehat{p}_n, \widehat{V}_n \right\rangle - \gamma \langle p, Y \rangle\right) \tag{25}$$

$$= (\widehat{R}_n - \mu) + \gamma\left(\left\langle (\widehat{p}_n - p), \widehat{V}_n \right\rangle + \left\langle p, (\widehat{V} - V) \right\rangle\right) \tag{26}$$

Therefore,

$$\Rightarrow \left|\mathbb{E}[\widehat{Q}_n] - \big(\mu + \gamma \langle p, V \rangle\big)\right| \leq \left|\mathbb{E}[(\widehat{R}_n - \mu)]\right| + \gamma\left(|\mathbb{E}[\widehat{p}_n - p]| \left|\widehat{V}_n\right| + p \left|\mathbb{E}[\widehat{V} - V]\right|\right) \tag{27}$$

$$\Rightarrow \left|\mathbb{E}[\widehat{Q}_n] - \big(\mu + \gamma \langle p, V \rangle\big)\right| \leq \left|\mathbb{E}[(\widehat{R}_n - \mu)]\right| + \gamma\left(L\,|\mathbb{E}[\widehat{p}_n - p]| + p \left|\mathbb{E}[\widehat{V} - V]\right|\right) \tag{28}$$

Also because $\lim_{n\to\infty} \mathbb{E}[\widehat{V}_{m,n}] = V_m$, $\lim_{n\to\infty} \frac{\widehat{N}_m^n}{n} = p_m$, and $\mathbb{E}[(\widehat{R}_n - \mu)] = 0$ so that,

$$\lim_{n\to\infty} \mathbb{E}[\widehat{Q}_n] = \mu + \gamma \sum_{m=1}^{M} p_m V_m.$$

That mean

$$\operatorname*{plim}_{n\to\infty} \widehat{Q}_n = \mu + \gamma \sum_{m=1}^{M} p_m V_m,$$

which concludes the proof. □

Results from Lemma C.1 is important as it shows the concentration for the Q value estimation given the concentration of V value of the children nodes.

Lemma C.2. *Let consider non-negative variables $x, y \in \mathbb{R}^+$, and a constant m that $0 \leq m \leq 1$. Then*

$$(x + y)^m \leq x^m + y^m. \tag{29}$$

We use Minkowski's inequality as shown below

Lemma C.3. ***(Minkowski's inequality)*** *Given $p \geq 1, \{x_i, y_i\} \in \mathbb{R}, i = 1, 2, ..., n$, then we have the following inequality*

$$\left(\sum_i (|x_i + y_i|)^p\right)^{\frac{1}{p}} \leq \left(\sum_i (|x_i|)^p\right)^{\frac{1}{p}} + \left(\sum_i (|y_i|)^p\right)^{\frac{1}{p}}. \tag{30}$$

Proof. This is a basic result. □

Lemma C.4. ***(Markov's inequality)*** *If X is a nonnegative random variable and $a > 0$, then the probability that X is at least a is at most the expectation of X divided by a:*

$$\mathbb{P}(X > a) \leq \frac{\mathbb{E}[X]}{a}. \tag{31}$$

Proof. This is a well-known result. □

D Convergence of CATSO and PATSO in Non-stationary multi-armed bandits

We note that in an MCTS tree, each node is considered a non-stationary multi-armed bandit where the average mean drifts due to the given action selection strategy. Therefore, we first study the convergence of CATSO and PATSO in non-stationary multi-armed bandits where the action selection is Thompson sampling, with the power mean backup operator at the root node. Detailed descriptions of the CATSO and PATSO in Non-stationary multi-armed bandits settings can be found in the main article in the Theoretical Analysis section.

We first establish the convergence and concentration properties for the power mean backup operator in non-stationary bandits, detailed in Theorem **??** for CATSO and Theorem **??** for PATSO.

To achieve these results, we demonstrate that the expected payoff of the power mean backup operator decays polynomially at a rate of $O(\frac{\log n}{n})$. This is supported by Lemma D.3 for `CATSO` and Lemma D.4 for `PATSO`. Critical to this analysis are Lemma D.1 and Lemma D.2, which establish an upper bound of $\log(n)$ for the expected number of suboptimal arm pulls.

We introduce some important definitions. F_a^n represents the empirical cumulative distribution function of arm a after n visitations, and F_a represents the cumulative distribution function of arm a. We employ the following distance measure: If P and Q are two distributions characterized by parameters $p = (p_0, p_1, \cdots, p_N)$ and $q = (q_0, q_1, \cdots, q_N)$ respectively, then the distance is defined as

$$d(P, Q) := \| p - q \|_\infty = \sup_{i \in [0,N]} |p_i - q_i|$$

This represents the L^∞ distance between p and q in $\mathbb{R}^{N+1}$. We also denotes $\mathrm{KL}(P \parallel Q)$ as the Kullback–Leibler divergence between P and Q, and denote $\mathcal{K}_{\inf}(F_a, \mu_\star) = \inf_{G:\mathbb{E}[G]>\mu_\star} \mathrm{KL}(F_a \parallel G)$. In addition, we denote $\mathcal{K}_{\inf}^{(N)}(F_a, \mu_\star) = \inf \left\{ \mathrm{KL}(F_a \parallel G) \middle| \text{the support of G} \in \left\{0, \frac{R(t)}{N}, \frac{2R(t)}{N}, \cdots, R(t)\right\}, \mathbb{E}[G] > \mu_\star \right\}$.

We see that the definition of $\mathcal{K}_{\inf}(F_a, \mu_\star)$ and $\mathcal{K}_{\inf}^{(N)}(F_a, \mu_\star)$ is only difference in the support set.

We denote the true parameter of arm a by $p_a = (p_a^0, p_a^1, \ldots, p_a^N)$ with $p_a^i = \mathbb{P}_{X \sim F_a}[X = \frac{i}{N}]$. We denote the parameter of the posterior distribution of arm a as $\alpha_a = (\alpha_a^0, \alpha_a^1, \ldots, \alpha_a^N)$. Since each arm a is non-stationary, we also denote the parameter of arm a after n visitations by $p_a(n) = (p_a^0(n), p_a^1(n), \ldots, p_a^N(n))$ with $p_a^i(n) = \mathbb{P}_{X \sim F_a^n}[X = \frac{i}{N}]$. The parameter of the posterior distribution of arm a denoted as $\alpha_a(n) = (\alpha_a^0(n), \alpha_a^1(n), \ldots, \alpha_a^N(n))$ We first show the results of an important Lemma D.1. The proof follows closely to the Proof of Proposition 7 [Riou and Honda, 2020]. The only difference is that in our settings, we study non-stationary bandits.

Lemma D.1. *Assume a non-stationary bandit setting satisfies Assumption 5.2, and that the `CATSO` (Categorical Thompson Sampling with Optimistic Bonus) algorithm is applied. Let $T_a(n)$ denote the number of times arm a is pulled by time step n. If a indexes a suboptimal arm, then for any $\epsilon_0 \geq 0$, the expected count of arm a satisfies*

$$\mathbb{E}[T_a(n)] \;\leq\; \frac{(1+\epsilon_0)\,\log n}{\mathcal{K}_{\inf}^{(N)}(F_a, \mu_\star)} \;+ \mathcal{O}(\sqrt{n}).$$

Proof. We have $\overline{\theta}_a(t) = [0, \frac{R(t)}{N}, \frac{2R(t)}{N}, \cdots, R(t)]^\top L_{a,t}$, with $L_{a,t} \sim \mathrm{Dir}(\alpha_a^0(t), \ldots, \alpha_a^N(t))$.

To analyze the expectation associated with selecting a suboptimal arm a, we follow the approach of [Riou and Honda, 2020] and decompose the term into two components:

$$\mathbb{E}\left[\sum_{t=1}^{n} \mathbb{1}(I(t) = a)\right] = \underbrace{\mathbb{E}\left[\sum_{t=1}^{n} \mathbb{1}(I(t) = a), \overline{\theta}_a(t) \geq \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}, d(\widehat{F}_{I(t)}, F_{I(t)}) \leq \epsilon_2)\right]}_{A1} \tag{32}$$

$$+ \underbrace{\mathbb{E}\left[\sum_{t=1}^{n} \mathbb{1}\left(I(t) = a, \left\{\overline{\theta}_a(t) < \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}\right\} \cup \left\{d(\widehat{F}_{I(t)}, F_{I(t)}) > \epsilon_2\right\}\right)\right]}_{A2} \tag{33}$$

<u>We first find an upper bound for A_1:</u>

$$A1 = \sum_{t=1}^{n}\sum_{m=1}^{n} \mathbb{E}\left[\mathbb{1}\left(I(t) = a, \overline{\theta}_a(t) \geq \mu_\star - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}; \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m\right)\right] \tag{34}$$

We first note that if the event

$$\left\{I(t) = a, \overline{\theta}_a(t) \geq \mu_\star - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}; \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m\right\} \tag{35}$$

occurs at step t for a certain $m \in [1, n]$, then $T_a(t') > T_a(t) = m$ for any $t' > t$.

Thus, for any $m \in [n]$

$$\sum_{t=1}^{n} \mathbb{1}\left(I(t) = a, \overline{\theta}_a(t) \geq \mu_\star - \epsilon_1; \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right) \leq 1. \quad (36)$$

We deduce that for any $m_0 \in [n]$,

$$A1 \leq m_0 + \sum_{t=1}^{n} \sum_{m=m_0}^{n} \mathbb{E}\left[\mathbb{1}\left(I(t) = a, \overline{\theta}_a(t) \geq \mu_\star - C \frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}; \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right) \right] \quad (37)$$

$$\leq m_0 + \sum_{t=1}^{n} \sum_{m=m_0}^{n} \mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - C \frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}; \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right) \quad (38)$$

$$\leq m_0 + \sum_{t=1}^{n} \sum_{m=m_0}^{n} \mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - C \frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}} \middle| \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right) \quad (39)$$

$$\times \mathbb{P}\left(\| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right). \quad (40)$$

For any fixed $\epsilon_1 > 0$, if we choose m_0 such that

$$m_0 \geq \frac{C}{\epsilon_1} \sqrt{n},$$

then

$$A1 \leq m_0 + \sum_{t=1}^{n} \sum_{m=m_0}^{n} \mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right) \quad (41)$$

$$\times \mathbb{P}\left(\| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right). \quad (42)$$

By applying the results of Lemma 13, Appendix F [Riou and Honda, 2020], we have

$$\mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| \alpha_a(t), T_a(t) = m \right) \leq C_0 (m+N+1)^{N/2} \exp\{-(m+N+1)\mathrm{KL}(P_{\alpha_a(t)} \| P^*_{\mu_\star - \epsilon_1})\}$$

where

$$P^*_{\mu_\star - \epsilon_1} = \arg \min_{x : u^\top x \geq \mu_\star - \epsilon_1} \mathrm{KL}(P_{\alpha_a} \| x) \quad \text{and} \quad P_{\alpha_a(t)} = \frac{1}{n + N + 1} \alpha_a(t).$$

By definition, we have $\mathrm{KL}(P_{\alpha_a(t)} \| P^*_{\mu_\star - \epsilon_1}) = \mathcal{K}_{\inf}(P_{\alpha_a(t)}, \mu_\star - \epsilon_1)$, and

$$\mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| \alpha_a(t), T_a(t) = m \right) \leq C_0 (m+N+1)^{N/2} \exp\{-(m+N+1)\mathcal{K}_{\inf}(P_{\alpha_a(t)}, \mu_\star - \epsilon_1)\},$$

where $C_0 = \frac{\exp\{1/12\}}{\Gamma(N+1)} \left(\frac{1}{\sqrt{2\pi}} \right)^N$.

On the other hand, $\mathcal{K}_{\inf}(x, \mu_\star - \epsilon_1)$ is continuous in $x \in [0, 1]^{N+1}$ on the probability simplex with respect to the L^∞ distance from ([Honda and Takemura, 2010], Theorem 7) and Lemma 18 in Appendix H [Riou and Honda, 2020]. Therefore, for any $\epsilon_3 > 0$, there exists $\epsilon_2 > 0$ and constant $C' > 0$ such that

$$\mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| \| \frac{\alpha_a(t)}{T_a(t) + N + 1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m \right) \quad (43)$$

$$\leq C' \exp\{-(m + N + 1)(\mathcal{K}_{\inf}(p_a, \mu_\star - \epsilon_1) - \epsilon_3)\} \quad (44)$$

Note that $\mathbb{P}\left(\| \frac{\alpha_a(t)}{T_a(t)+N+1} - p_a(t) \|_\infty \leq \epsilon_2, T_a(t) = m\right) \leq 1$. We have

$$A1 \leq m_0 + C_1' \sum_{t=1}^{n} \exp\{-(m_0+N+1)(\mathcal{K}_{\text{inf}}(p_a, \mu_\star - \epsilon_1) - \epsilon_3)\} \tag{45}$$

$$\leq m_0 + C_1' n \exp\{-(m_0+N+1)(\mathcal{K}_{\text{inf}}(p_a, \mu_\star - \epsilon_1) - \epsilon_3)\} \tag{46}$$

Furthermore, as from ([Honda and Takemura, 2010], Theorem 7), it is proven that $\mu \to \mathcal{K}_{\text{inf}}(F, \mu)$ is continuous for $\mu < 1$, when we scale reward from [0,1] to $[0, R]$ therefore μ from [0,1] to $[0, R]$. We have $\mu \to \mathcal{K}_{\text{inf}}(F, \mu)$ is continuous for $\mu < R$. Therefore, $\forall \epsilon_4 > 0, \exists \epsilon_1 > 0$, such that

$$|\mathcal{K}_{\text{inf}}(p_a, \mu^* - \epsilon_1) - \mathcal{K}_{\text{inf}}(p_a, \mu^*)| \leq \epsilon_4$$

which imples that for any ϵ_0, there exists ϵ_1, ϵ_2 such that

$$\mathcal{K}_{\text{inf}}(p_a, \mu^* - \epsilon_1) - \epsilon_3 \geq \frac{1}{1+\epsilon_0} \mathcal{K}_{\text{inf}}(p_a, \mu^*).$$

Thus, if we choose

$$\begin{aligned} m_0 &= \max\left\{\frac{C}{\epsilon_1}\sqrt{n}, \frac{\log n}{\mathcal{K}_{\text{inf}}(p_a, \mu_\star - \epsilon_1) - \epsilon_3} - N - 1\right\} \\ &\leq \mathcal{O}(\sqrt{n}) + \frac{(1+\epsilon_0)\log n}{\mathcal{K}_{\text{inf}}^{(N)}(F_a, \mu_\star)}. \end{aligned}$$

then $A1 \leq m_0 + C_1'$.

<u>Upper bound for A_2:</u>

To provide an upper bound for A_2, we decompose the term into two parts:

$$\begin{aligned} A_2 = &\underbrace{\mathbb{E}\left[\sum_{t=1}^{n} \nVdash\left(I(t) = a, \left\{\overline{\theta}_a(t) < \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}\right\}\right)\right]}_{R1} \\ &+ \underbrace{\mathbb{E}\left[\sum_{t=1}^{n} \nVdash\left(I(t) = a, \left\{d(\widehat{F}_{I(t)}, F_{I(t)}) > \epsilon_2\right\}\right)\right]}_{R2} \end{aligned}$$

According to Proposition 8 [Riou and Honda, 2020], for any $\epsilon_2 > 0$, we have

$$R2 \leq O(1). \tag{47}$$

To bound B_1, we note that

$$R1 \leq \mathbb{E}\left[\sum_{t=1}^{n} \nVdash\left(\overline{\theta}_a(t) < \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}\right)\right]$$

We decompose the set

$$\{1, 2, \ldots, n\} = \bigcup V_x, \quad \text{where} \quad V_x = \left\{t : x \leq \frac{T_a(t)}{\sqrt{t}} < x+1\right\}, \quad x \in \{0, 1, \ldots, \sqrt{n}\}$$

We have

$$\begin{aligned} \mathbb{E}\left[\sum_{t \in V_x} \nVdash\left(\overline{\theta}_a(t) < \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}\right)\right] &\leq \mathbb{E}\left[\sum_{t \in V_x}^{n} \nVdash\left(\overline{\theta}_a(t) < \mu_* - \frac{C}{\sqrt{x+1}}\right)\right] \\ &\leq \mathbb{E}\left[\sum_{t=1}^{n} \nVdash\left(\overline{\theta}_a(t) < \mu_* - \frac{C}{\sqrt{x+1}}\right)\right] = \mathcal{O}(1) \end{aligned}$$

by Proposition 8 [Riou and Honda, 2020]. Since there is at most $\sqrt{n}+1$ such non-empty V_x, we deduce that

$$R1 = \mathcal{O}(\sqrt{n}).$$

By combining the upper bounds on A_1, R_1 and R_2, we have

$$\mathbb{E}[T_a(n)] \leq \frac{(1+\epsilon_0)\log n}{\mathcal{K}_{\inf}^{(N)}(F_a, \mu_\star)} + \mathcal{O}(\sqrt{n}). \tag{48}$$

This completes the proof. □

Lemma D.2. *Suppose the payoff sequence satisfies Assumption 5.2, and consider a non-stationary bandit setting in which Particle Thompson Sampling with Optimistic Bonus (`PATSO`) is employed. Let $T_a(n)$ denote the number of times a suboptimal arm a is played up to time n. For any $\epsilon_0 \geq 0$, the expected number of pulls of each suboptimal arm a satisfies*

$$\mathbb{E}\big[T_a(n)\big] \;\leq\; \frac{\log n}{\mathcal{K}_{\inf}\big(F_a, \mu_\star\big) \;-\; \epsilon_0} + \mathcal{O}(\sqrt{n}).$$

where $\mu_\star$ is the (unique) optimal mean reward, F_a is the distribution associated with arm a.

Proof. In this Theorem, we use the Levy distance. Recall that the Levy distance between two cumulative distribution functions F and G on $[0,1]$ is defined as

$$D_L(F,G) = \inf\{\epsilon > 0 : \forall x \in [0,1], F(x-\epsilon) - \epsilon \leq G(x) \leq F(x+\epsilon) + \epsilon\}.$$

The proof follows the same steps as in Lemma D.1. We derive

$$\mathbb{E}\left[\sum_{t=1}^{n} \mathbb{1}(I(t)=a)\right] = \underbrace{\mathbb{E}\left[\sum_{t=1}^{n} \mathbb{1}(I(t)=a), \overline{\phi}_{a,t} \geq \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}, D_L(\widehat{F}_{I(t)}, F_{I(t)}) \leq \epsilon_2)\right]}_{B1} \tag{49}$$

$$+\underbrace{\mathbb{E}\left[\sum_{t=1}^{n} \mathbb{1}(I(t)=a), \left\{\overline{\phi}_{a,t} < \mu_* - C\,\frac{t^{\frac{1}{4}}}{T_a(t)^{\frac{1}{2}}}\right\} \cup \left\{D_L(\widehat{F}_{I(t)}, F_{I(t)}) > \epsilon_2\right\}\right]}_{B2} \tag{50}$$

As in Lemma D.1, when for any fixed $\epsilon_1 > 0$, if we choose m_0 such that

$$m_0 \geq \frac{C}{\epsilon_1}\sqrt{n},$$

then

$$B1 \leq m_0 + \sum_{t=1}^{n}\sum_{m=m_0}^{n} \mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| D_L\left(\widehat{F}_a(t), F_a(t)\right) \leq \epsilon_2, T_a(t) = m\right) \tag{51}$$

$$\times\, \mathbb{P}\left(D_L\left(\widehat{F}_a(t), F_a(t)\right) \leq \epsilon_2, T_a(t) = m\right) \tag{52}$$

According to Lemma 15 in Appendix G.1 [Riou and Honda, 2020] on conditional probabilities, for any $\nu \in (0,1)$ we have

$$\mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| D_L\left(\widehat{F}_a(t), F_a(t)\right) \leq \epsilon_2, T_a(t) = m\right) \tag{53}$$

$$\leq \frac{1}{\nu}\exp\left\{-m\left(\mathcal{K}_{\inf}(\widehat{F}_a(t), \mu_\star - \epsilon_1) - \nu\frac{\mu_\star - \epsilon_1}{1-(\mu_\star - \epsilon_1)}\right)\right\} \tag{54}$$

Because $\mathcal{K}_{\inf}(F,\mu)$ is continuous in F with respect to the Levy distance from [Honda and Takemura, 2010], Theorem 7, for any $\epsilon_3 > 0$ there exists $\epsilon_2 > 0$ such that

$$D_L(\widehat{F}_a(t), F_a) \leq \epsilon_2 \Rightarrow \left|\mathcal{K}_{\inf}(\widehat{F}_a(t), \mu_\star - \epsilon_1) - \mathcal{K}_{\inf}(F_a, \mu_\star - \epsilon_1)\right| \leq \epsilon_3 \tag{55}$$

Therefore, $\forall \nu \in (0,1)$ and for any $\epsilon_5 > 0$, there exists $\epsilon_1, \epsilon_2 > 0$ such that

$$\mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| D_L\left(\widehat{F}_a(t), F_a(t)\right) \leq \epsilon_2, T_a(t) = m\right) \tag{56}$$

$$\leq \frac{1}{\nu}\left(-m\left(\mathcal{K}_{\inf}(F_a, \mu_\star - \epsilon_1) - \epsilon_3 - \nu\frac{\mu_\star - \epsilon_1}{1 - (\mu_\star - \epsilon_1)}\right)\right) \tag{57}$$

$$\overset{\text{(Theorem 6 [Honda and Takemura, 2010])}}{\leq} \frac{1}{\nu}\left(-m\left(\mathcal{K}_{\inf}(F_a, \mu_\star) - \frac{\epsilon_1}{1-\mu_\star} - \epsilon_3 - \nu\frac{\mu_\star - \epsilon_1}{1 - (\mu_\star - \epsilon_1)}\right)\right) \tag{58}$$

This implies that $\forall \epsilon_0 > 0$, there exists $\nu \in (0,1), \epsilon_1 > 0$ and $\epsilon_2 > 0$ such that

$$\mathbb{P}\left(\overline{\theta}_a(t) \geq \mu_\star - \epsilon_1 \middle| D_L\left(\widehat{F}_a(t), F_a(t)\right) \leq \epsilon_2, T_a(t) = m\right) \leq \frac{1}{\nu}\exp\left\{-m(\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0)\right\} \tag{59}$$

Therefore, according to inequality (52) and the fact that

$$\mathbb{P}\left(D_L\left(\widehat{F}_a(t), F_a(t)\right) \leq \epsilon_2, T_a(t) = m\right) \leq 1,$$

we have

$$B1 \leq m_0 + \sum_{t=1}^{n}\frac{1}{\nu}\exp\left\{-m_0(\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0)\right\} \leq m_0 + \frac{1}{\nu}T\exp\left\{-m_0(\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0)\right\}. \tag{60}$$

Choose

$$m_0 = \max\left\{\frac{C}{\epsilon_1}\sqrt{n}, \frac{\log n}{\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0}\right\} \leq \mathcal{O}(\sqrt{n}) + \frac{\log n}{\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0}.$$

we have

$$B1 \leq \frac{\log n}{\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0} + \mathcal{O}(\sqrt{n}). \tag{61}$$

Using the same argument as in the proof of Lemma D.1, we have $B2 \leq \mathcal{O}(\sqrt{n})$.

That leads us to

$$\mathbb{E}[T_a(n)] \leq \frac{\log n}{\mathcal{K}_{\inf}(F_a, \mu_\star) - \epsilon_0} + \mathcal{O}(\sqrt{n}), \tag{62}$$

which completes the proof. □

Lemma D.3. *Let* `CATSO` *(Categorical Thompson Sampling with Optimistic Bonus) be applied to a non-stationary bandit problem whose payoff sequence satisfies Assumption 5.2. Define the power-mean estimator*

$$\widehat{\mu}_n(p) \;=\; \left(\sum_{a=1}^{K}\frac{T_a(n)}{n}\left[\widehat{\mu}_{a,T_a(n)}\right]^p\right)^{\frac{1}{p}},$$

and let $\delta_{\star,n} = \mu_\star - \mu_{\star,n}$. For any $p \geq 1$ and $\epsilon_0 > 0$, the following upper bound on $\left|\mathbb{E}\big[\widehat{\mu}_n(p)\big] - \mu_\star\right|$ holds:

$$\left|\mathbb{E}\big[\widehat{\mu}_n(p)\big] \;-\; \mu_\star\right| \;\leq\; \left|\delta_{\star,n}\right| \;+\; \frac{R}{n}\sum_{\substack{a=1\\ a\neq a_*}}^{K}\left\{\frac{(1+\epsilon_0)\,\log n}{\mathcal{K}^{(N)}\big(F_a, \mu_\star\big)} \;+\; \mathcal{O}(\sqrt{n})\right\}.$$

Proof. We observe that

$$|\widehat{\mu}_n(p) - \mu_\star| \leq |\widehat{\mu}_n(p) - \mu_{\star,n}| + |\mu_\star - \mu_{\star,n}| = |\widehat{\mu}_n(p) - \mu_{\star,n}| + |\delta_{\star,n}| \tag{63}$$

Furthermore,

$$\widehat{\mu}_{a,T_a(n)} \leq \mu_{a,n} + \left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|. \tag{64}$$

Since $\mu_{\star,n} = \max_{a\in[K]}\{\mu_{a,n}\}$, we have

$$\widehat{\mu}_n(p) - \mu_{\star,n} = \widehat{\mu}_n(p) - \sum_{a=1}^{K} T_a(n)\mu_{\star,n} \leq \left(\sum_{a=1}^{K} \frac{T_a(n)}{n}\left(\widehat{\mu}_{a,T_a(n)}\right)^p\right)^{\frac{1}{p}} - \left(\sum_{a=1}^{K} \frac{T_a(n)}{n}\left(\mu_{a,n}\right)^p\right)^{\frac{1}{p}} \tag{65}$$

$$= \frac{\left(\sum_{a=1}^{K} T_a(n)\left(\widehat{\mu}_{a,T_a(n)}\right)^p\right)^{\frac{1}{p}} - \left(\sum_{a=1}^{K} T_a(n)\left(\mu_{a,n}\right)^p\right)^{\frac{1}{p}}}{n^{\frac{1}{p}}} \tag{66}$$

Applying Minkowski's inequality from Lemma C.3, and the result of (64), we have

$$\widehat{\mu}_n(p) - \mu_{\star,n} \leq \frac{\left(\sum_{a=1}^{K} T_a(n)\left(\mu_a + \left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|\right)^p\right)^{\frac{1}{p}} - \left(\sum_{a=1}^{K} T_a(n)\left(\mu_{a,n}\right)^p\right)^{\frac{1}{p}}}{n^{\frac{1}{p}}} \tag{67}$$

$$\leq \frac{\left(\sum_{a=1}^{K} T_a(n)\left(\left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|\right)^p\right)^{\frac{1}{p}}}{n^{\frac{1}{p}}} \tag{68}$$

On the other hand,

$$\mu_{\star,n} - \widehat{\mu}_n(p) = \frac{n\mu_{\star,n} - n\widehat{\mu}_n(p)}{n} = \frac{n\mu_{\star,n} - \left(\sum_{a=1}^{K} T_a(n)\mu_{a,n}\right) + \sum_{a=1}^{K} T_a(n)\mu_{a,n} - n\widehat{\mu}_n(p)}{n} \tag{69}$$

$$= \frac{\sum_{a=1,a\neq a_*}^{K} T_a(n)\left|\mu_{\star,n} - \mu_{a,n}\right| + \sum_{a=1}^{K} T_a(n)\mu_{a,n} - n\widehat{\mu}_n(p)}{n} \tag{70}$$

$$\leq R\sum_{a=1,a\neq a_*}^{K} \frac{T_a(n)}{n} + \sum_{a=1}^{K} \frac{T_a(n)}{n}\mu_{a,n} - \widehat{\mu}_n(p) \tag{71}$$

Because power mean is an increasing function of p, so that

$$\sum_{a=1}^{K} \frac{T_a(n)}{n}\mu_{a,n} \leq \left(\sum_{a=1}^{K} \frac{T_a(n)}{n}\left(\mu_{a,n}\right)^p\right)^{1/p}.$$

Furthermore, we observe that

$$\mu_{a,n} \leq \widehat{\mu}_{a,T_a(n)} + \left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|.$$

So that, from equation (71) we have

$$\mu_{\star,n} - \widehat{\mu}_n(p) \leq R \sum_{a=1, a\neq a_*}^{K} \frac{T_a(n)}{n} + \left(\sum_{a=1}^{K} \frac{T_a(n)}{n} \left(\mu_{a,n}\right)^p \right)^{1/p} - \widehat{\mu}_n(p) \tag{72}$$

$$\leq R \sum_{a=1, a\neq a_*}^{K} \frac{T_a(n)}{n} \tag{73}$$

$$+ \frac{\left(\sum_{a=1}^{K} T_a(n) \left(\widehat{\mu}_{a,T_a(n)} + \left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|\right)^p\right)^{\frac{1}{p}} - \left(\sum_{a=1}^{K} T_a(n) \left(\widehat{\mu}_{a,T_a(n)}\right)^p\right)^{\frac{1}{p}}}{n^{\frac{1}{p}}} \tag{74}$$

$$\overset{\text{(Minkovski's inequality)}}{\leq} R \sum_{a=1, a\neq a_*}^{K} \frac{T_a(n)}{n} + \frac{\left(\sum_{a=1}^{K} T_a(n) \left(\left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|\right)^p\right)^{\frac{1}{p}}}{n^{\frac{1}{p}}} \tag{75}$$

$$\overset{\text{(Properties of } L^p \text{ norm)}}{\leq} R \sum_{a=1, a\neq a_*}^{K} \frac{T_a(n)}{n} + \frac{\left(\sum_{a=1}^{K} T_a(n) \left(\left|\widehat{\mu}_{a,T_a(n)} - \mu_{a,n}\right|\right)\right)}{n^{\frac{1}{p}}} \tag{76}$$

$$= R \sum_{a=1, a\neq a_*}^{K} \frac{T_a(n)}{n} + \frac{\sum_{a=1}^{K} \left(\left|\sum_{t}^{T_a(n)} R_{a,t} - T_a(n)\mu_{a,n}\right|\right)}{n^{\frac{1}{p}}} \tag{77}$$

Therefore

$$|\mathbb{E}[\widehat{\mu}_n(p) - \mu_{\star,n}]| \leq R \sum_{a=1, a\neq a_*}^{K} \frac{\mathbb{E}[T_a(n)]}{n} + \frac{\mathbb{E}\left[\left(\left|\sum_{a=1}^{K} \sum_{t}^{T_a(n)} R_{a,t} - T_a(n)\mu_{a,n}\right|\right)\right]}{n^{\frac{1}{p}}} \tag{78}$$

$$\leq R \sum_{a=1, a\neq a_*}^{K} \frac{\mathbb{E}[T_a(n)]}{n} + O(n^{-1/2}) \tag{79}$$

Please note that because we study non-stationary bandits, $\mathbb{E}[\sum_{t}^{T_a(n)} R_{a,t}] = T_a(n)\mu_{a,n}$ conditionally on $T_a(n)$. By Jensen's inequality and the law of total expectation, the second term is bounded by:

$$\frac{\mathbb{E}\left[\left(\sum_{a=1}^{K} \left|\sum_{t}^{T_a(n)} R_{a,t} - T_a(n)\mu_{a,n}\right|\right)\right]}{n^{\frac{1}{p}}} \leq \frac{\sum_{a=1}^{K} \mathbb{E}[\sqrt{T_a(n)} \cdot R]}{n^{1/p}} \leq \frac{KR\sqrt{n}}{n^{1/p}} = O(n^{-1/2})$$

for $p \geq 1$, where we used $\mathbb{E}[|\bar{X}_m - \mu|] \leq R/\sqrt{m}$ by Jensen applied to the variance bound. Since this term is $O(n^{-1/2})$, it is absorbed into the final rate.

According to Lemma D.1, we have

$$|\mathbb{E}[\widehat{\mu}_n(p) - \mu_{\star,n}]| \leq R \sum_{a=1, a\neq a_*}^{K} \frac{\mathbb{E}[T_a(n)]}{n} \leq \frac{R}{n} \sum_{a=1, a\neq a_*}^{K} \left\{ \frac{(1+\epsilon_0)\log n}{\mathcal{K}^{(N)}(F_a, \mu_\star)} + \mathcal{O}(\sqrt{n}) \right\}, \tag{80}$$

which concludes the proof. □

Lemma D.4. *Let* `PATSO` *(Particle Thompson Sampling with Optimistic Bonus) be applied to a non-stationary bandit problem whose payoff process satisfies Assumption 5.2. Define the power-mean estimator*

$$\widehat{\mu}_n(p) \;=\; \left(\sum_{a=1}^{K} \frac{T_a(n)}{n} \left[\widehat{\mu}_{a,T_a(n)}\right]^p \right)^{1/p},$$

and let $\delta_{\star,n} = \mu_\star - \mu_{\star,n}$. For any $p \geq 1$ and $\epsilon_0 > 0$, the following holds:

$$\left|\mathbb{E}\big[\widehat{\mu}_n(p)\big] \;-\; \mu_\star\right| \;\leq\; \left|\delta_{\star,n}\right| \;+\; \frac{R}{n} \sum_{\substack{a=1 \\ a\neq a_*}}^{K} \Big\{ \frac{\log n}{\mathcal{K}_{\inf}\big(F_a, \mu_\star\big) - \epsilon_0} \;+\; \mathcal{O}(\sqrt{n}) \Big\}.$$

Proof. Similar to Lemma D.3, we can derive

$$|\mathbb{E}[\widehat{\mu}_n(p) - \mu_{\star,n}]| \le |\delta_{\star,n}| + R \sum_{a=1, a\neq a_*}^{K} \frac{\mathbb{E}[T_a(n)]}{n}. \tag{81}$$

and the bound is obtained as a direct corollary of Lemma D.2. □

Definition D.5. We write

$$\operatorname*{plim}_{n\to\infty} \widehat{V}_n = V$$

if, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\Big(|\widehat{V}_n - V| > \varepsilon\Big) = 0.$$

In words, $\widehat{V}_n$ converges to V in probability.

Definition D.6 (Polynomial concentration)**.** For $\alpha, \beta > 0$, we write

$$\widehat{X}_n \xrightarrow[\alpha,\beta]{n\to\infty} x$$

if there exists a constant $c > 0$ such that, for all $n \ge 1$ and all $\varepsilon > n^{-\alpha/\beta}$,

$$\Pr\Big(|\widehat{X}_n - x| > \varepsilon\Big) \le c\, n^{-\alpha} \varepsilon^{-\beta}.$$

Assumption D.7 (Arm-level concentration of PATSO maintained means)**.** For each arm $a \in [K]$ and each local sample size $m \ge 1$, let $\mathcal{G}_{a,m}$ be the σ-field generated by the first m observed returns from arm a. Let $\phi_{a,m}$ denote the PATSO Thompson draw after m pulls of arm a, and define the maintained mean by

$$\widehat{\mu}_{a,m} := \mathbb{E}[\phi_{a,m} \mid \mathcal{G}_{a,m}]\,.$$

Assume that there exist constants $\alpha, \beta, c_\mu > 0$ such that, for all $a \in [K]$, all $m \ge 1$, and all $\varepsilon > m^{-\alpha/\beta}$,

$$\Pr(|\widehat{\mu}_{a,m} - \mu_a| > \varepsilon) \le c_\mu\, m^{-\alpha} \varepsilon^{-\beta},$$

where

$$\mu_a := \lim_{m\to\infty} \mathbb{E}[\widehat{\mu}_{a,m}].$$

Assumption D.8 (Conditional index tails)**.** There exists $c_{\mathrm{idx}} > 0$ such that for every round $t \ge K$, every arm $c \in [K]$, every $m \in \{1, \dots, t\}$, and every history $\mathcal{F}_t$ with $T_c(t) = m$,

$$\Pr\left(\phi_{c,m}^{(t)} - \mu_c > C\, \frac{t^{b/\beta}}{m^{\alpha/\beta}} \;\middle|\; \mathcal{F}_t\right) \le c_{\mathrm{idx}}\, C^{-\beta} t^{-b},$$

and

$$\Pr\left(\mu_c - \phi_{c,m}^{(t)} > C\, \frac{t^{b/\beta}}{m^{\alpha/\beta}} \;\middle|\; \mathcal{F}_t\right) \le c_{\mathrm{idx}}\, C^{-\beta} t^{-b}.$$

Lemma D.9 (Suboptimal pull tail under conditional index tails)**.** *Assume Assumption D.8. Let*

$$\mu_\star := \max_{a\in[K]} \mu_a, \qquad a_\star \in \arg\max_{a\in[K]} \mu_a,$$

and assume $a_\star$ is unique. For $a \neq a_\star$, define

$$\Delta_a := \mu_\star - \mu_a, \qquad A_a(n) := \left\lceil \left(\frac{2C}{\Delta_a}\right)^{\beta/\alpha} n^{b/\alpha} \right\rceil.$$

If each arm is sampled once initially and for all $t \ge K$,

$$A_{t+1} \in \arg\max_{c\in[K]} \left\{ \phi_{c,T_c(t)}^{(t)} + C\, \frac{t^{b/\beta}}{T_c(t)^{\alpha/\beta}} \right\},$$

then for every suboptimal arm $a \neq a_\star$ and every integer $u \ge A_a(n)$,

$$\Pr(T_a(n) > u) \le \frac{2 c_{\mathrm{idx}}\, C^{-\beta}}{b-1}\, (u-1)^{-(b-1)}.$$

Proof. Fix a suboptimal arm $a \neq a_\star$ and an integer $u \geq A_a(n)$. For each $t \in \{u, \ldots, n-1\}$, define

$$E_t := \{T_a(t) = u,\ A_{t+1} = a\}.$$

These events are disjoint, and

$$\{T_a(n) > u\} = \bigcup_{t=u}^{n-1} E_t.$$

Hence

$$\Pr(T_a(n) > u) = \sum_{t=u}^{n-1} \Pr(E_t).$$

Now fix $t \in \{u, \ldots, n-1\}$. On E_t, let

$$s := T_{a_\star}(t).$$

Since arm a is selected at round $t+1$, its optimistic index is at least that of the optimal arm:

$$U_{a,t,u}^{(t)} \geq U_{a_\star,t,s}^{(t)},$$

where

$$U_{c,t,m}^{(t)} := \phi_{c,m}^{(t)} + C\,\frac{t^{b/\beta}}{m^{\alpha/\beta}}.$$

Therefore, on E_t, it is impossible that both

$$U_{a,t,u}^{(t)} \leq \mu_\star \qquad \text{and} \qquad U_{a_\star,t,s}^{(t)} > \mu_\star$$

hold simultaneously. Thus

$$E_t \subseteq \{U_{a,t,u}^{(t)} > \mu_\star\} \cup \{U_{a_\star,t,T_{a_\star}(t)}^{(t)} \leq \mu_\star\}.$$

Because $u \geq A_a(n)$ and $t \leq n$, we also have $u \geq A_a(t)$, so

$$C\,\frac{t^{b/\beta}}{u^{\alpha/\beta}} \leq \frac{\Delta_a}{2}.$$

Hence, on $\{T_a(t) = u\}$,

$$\{U_{a,t,u}^{(t)} > \mu_\star\} \subseteq \left\{\phi_{a,u}^{(t)} - \mu_a > C\,\frac{t^{b/\beta}}{u^{\alpha/\beta}}\right\}.$$

Using Assumption D.8,

$$\Pr\Big(E_t \cap \{U_{a,t,u}^{(t)} > \mu_\star\}\Big) \leq c_{\text{idx}}\, C^{-\beta} t^{-b}.$$

Similarly,

$$\{U_{a_\star,t,T_{a_\star}(t)}^{(t)} \leq \mu_\star\} = \left\{\mu_\star - \phi_{a_\star,T_{a_\star}(t)}^{(t)} > C\,\frac{t^{b/\beta}}{T_{a_\star}(t)^{\alpha/\beta}}\right\},$$

so again by Assumption D.8,

$$\Pr\Big(E_t \cap \{U_{a_\star,t,T_{a_\star}(t)}^{(t)} \leq \mu_\star\}\Big) \leq c_{\text{idx}}\, C^{-\beta} t^{-b}.$$

Combining the two bounds,

$$\Pr(E_t) \leq 2c_{\text{idx}}\, C^{-\beta} t^{-b}.$$

Summing over $t = u, \ldots, n-1$,

$$\Pr(T_a(n) > u) \leq 2c_{\text{idx}}\, C^{-\beta} \sum_{t=u}^{n-1} t^{-b} \leq \frac{2c_{\text{idx}}\, C^{-\beta}}{b-1}(u-1)^{-(b-1)}.$$

This completes the proof. □

Lemma D.10 (Reduction from weighted random-time control)**.** *Let $a_\star$ be the unique optimal arm and set*

$$\mu_\star := \mu_{a_\star}, \qquad \alpha' := (b-1)\left(1-\frac{b}{\alpha}\right), \qquad \beta' := b-1.$$

Assume the following two properties.

(i) Weighted random-time estimation bound. *There exist constants $C_B > 0$ and $c_B > 0$ such that, for all $n \geq 1$ and all*

$$x \geq c_B\, n^{-p(1-b/\alpha)},$$

$$\Pr\left(\sum_{a=1}^{K} \frac{T_a(n)}{n} \left|\widehat{\mu}_{a,T_a(n)} - \mu_a\right|^p > x\right) \leq C_B\, n^{-\alpha'} x^{-\beta'/p}.$$

(ii) Strict tail bound for suboptimal pulls. *For each suboptimal arm $a \neq a_\star$, there exists*

$$c_a := \left(\frac{2C}{\Delta_a}\right)^{\beta/\alpha}, \qquad \Delta_a := \mu_\star - \mu_a,$$

such that, for all integers

$$u \geq A_a(n) := \left\lceil c_a\, n^{b/\alpha} \right\rceil,$$

$$\Pr(T_a(n) > u) \leq \frac{\bar{c}_{\mathrm{TS}}}{b-1}\,(u-1)^{-\beta'}.$$

Let

$$c_\Delta := \max_{a \neq a_\star} c_a.$$

Then there exist constants $c_\varepsilon > 0$ and $C_{\mathrm{PATSO}} > 0$, depending only on

$$(K, p, R, C_B, \bar{c}_{\mathrm{TS}}, c_\Delta),$$

such that for all $n \geq 1$ and all

$$\varepsilon \geq c_\varepsilon\, n^{-(1-b/\alpha)},$$

we have

$$\Pr\left(\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\right| > \varepsilon\right) \leq C_{\mathrm{PATSO}}\, n^{-\alpha'} \varepsilon^{-\beta'}.$$

Proof. Define

$$w_{a,n} := \frac{T_a(n)}{n}, \qquad S_n := \sum_{a \neq a_\star} w_{a,n},$$

and

$$B_n := \left(\sum_{a=1}^{K} w_{a,n} \left|\widehat{\mu}_{a,T_a(n)} - \mu_a\right|^p\right)^{1/p}.$$

Also define the deterministic comparison quantity

$$M_n := \left(\sum_{a=1}^{K} w_{a,n} \mu_a^p\right)^{1/p}.$$

I. Deterministic decomposition.

Since

$$\widehat{\mu}_{a,T_a(n)} \leq \mu_a + \left|\widehat{\mu}_{a,T_a(n)} - \mu_a\right|,$$

Minkowski's inequality yields

$$\widehat{\mu}_n^{\mathrm{PATSO}}(p) \leq M_n + B_n.$$

Similarly,

$$\mu_a \leq \widehat{\mu}_{a,T_a(n)} + \left|\widehat{\mu}_{a,T_a(n)} - \mu_a\right|,$$

so again by Minkowski,

$$M_n \le \widehat{\mu}_n^{\mathrm{PATSO}}(p) + B_n.$$

Hence

$$\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - M_n\right| \le B_n.$$

Next, because $a_\star$ is the unique optimal arm and $0 \le \mu_a \le \mu_\star \le R$ for all a,

$$M_n \le \mu_\star.$$

Moreover,

$$M_n \ge (w_{a_\star,n}\mu_\star^p)^{1/p} = \mu_\star\, w_{a_\star,n}^{1/p} = \mu_\star(1 - S_n)^{1/p}.$$

Therefore

$$0 \le \mu_\star - M_n \le \mu_\star\Big(1 - (1 - S_n)^{1/p}\Big).$$

Since $p \ge 1$ and $s \mapsto (1-s)^{1/p}$ satisfies

$$(1-s)^{1/p} \ge 1 - s \qquad (0 \le s \le 1),$$

we get

$$0 \le \mu_\star - M_n \le \mu_\star S_n \le R S_n.$$

Combining the two bounds gives

$$\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\right| \le B_n + R S_n.$$

Hence, for every $\varepsilon > 0$,

$$\Pr\big(\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\right| > \varepsilon\big) \le \Pr\Big(B_n > \frac{\varepsilon}{2}\Big) + \Pr\Big(S_n > \frac{\varepsilon}{2R}\Big). \tag{82}$$

II. Tail bound for the suboptimal mass S_n.

Fix $y > 0$ and define

$$u_y := \left\lfloor \frac{yn}{K-1} \right\rfloor.$$

If $S_n > y$, then

$$\sum_{a \ne a_\star} T_a(n) > yn,$$

so by the pigeonhole principle there exists some suboptimal arm $a \ne a_\star$ such that

$$T_a(n) > \frac{yn}{K-1} \ge u_y.$$

Hence

$$\{S_n > y\} \subseteq \bigcup_{a \ne a_\star} \{T_a(n) > u_y\}. \tag{83}$$

Now choose

$$c_S := 4(K-1)(c_\Delta + 1).$$

Assume

$$y \ge c_S\, n^{-(1-b/\alpha)}.$$

Then

$$\frac{yn}{K-1} \ge 4(c_\Delta + 1)n^{b/\alpha} \ge 4,$$

and therefore

$$u_y - 1 \ge \frac{1}{2} \cdot \frac{yn}{K-1}.$$

Also,

$$u_y \ge \frac{yn}{K-1} - 1 \ge 4(c_\Delta + 1)n^{b/\alpha} - 1 \ge c_\Delta n^{b/\alpha} + 1 \ge A_a(n)$$

for every suboptimal arm $a \neq a_\star$. Thus, applying the strict pull-tail bound and (83),

$$\begin{aligned}\Pr(S_n > y) &\leq \sum_{a \neq a_\star} \Pr(T_a(n) > u_y) \\ &\leq \frac{(K-1)\bar{c}_{\mathrm{TS}}}{b-1}(u_y - 1)^{-\beta'} \\ &\leq \frac{2^{\beta'}(K-1)^{\beta'+1}\bar{c}_{\mathrm{TS}}}{b-1}\, n^{-\beta'} y^{-\beta'}.\end{aligned}$$

Since

$$\alpha' = (b-1)\left(1 - \frac{b}{\alpha}\right) \leq b - 1 = \beta',$$

we have $n^{-\beta'} \leq n^{-\alpha'}$ for all $n \geq 1$. Therefore,

$$\Pr(S_n > y) \leq C_S\, n^{-\alpha'} y^{-\beta'},$$

where

$$C_S := \frac{2^{\beta'}(K-1)^{\beta'+1}\bar{c}_{\mathrm{TS}}}{b-1}.$$

III. Bound the two terms in (82)**.**

First, if

$$\varepsilon \geq 2c_B^{1/p}\, n^{-(1-b/\alpha)},$$

then

$$\left(\frac{\varepsilon}{2}\right)^p \geq c_B\, n^{-p(1-b/\alpha)}.$$

Hence assumption (i) gives

$$\begin{aligned}\Pr\Big(B_n > \frac{\varepsilon}{2}\Big) &= \Pr\Big(B_n^p > \Big(\frac{\varepsilon}{2}\Big)^p\Big) \\ &\leq C_B\, n^{-\alpha'} \left(\left(\frac{\varepsilon}{2}\right)^p\right)^{-\beta'/p} \\ &= 2^{\beta'} C_B\, n^{-\alpha'} \varepsilon^{-\beta'}.\end{aligned}$$

Second, if

$$\varepsilon \geq 2Rc_S\, n^{-(1-b/\alpha)},$$

then

$$\frac{\varepsilon}{2R} \geq c_S\, n^{-(1-b/\alpha)},$$

so Step 2 gives

$$\begin{aligned}\Pr\Big(S_n > \frac{\varepsilon}{2R}\Big) &\leq C_S\, n^{-\alpha'} \left(\frac{\varepsilon}{2R}\right)^{-\beta'} \\ &= (2R)^{\beta'} C_S\, n^{-\alpha'} \varepsilon^{-\beta'}.\end{aligned}$$

Now choose

$$c_\varepsilon := \max\{2c_B^{1/p},\, 2Rc_S\}.$$

Then for every

$$\varepsilon \geq c_\varepsilon\, n^{-(1-b/\alpha)},$$

both bounds apply, and (82) yields

$$\Pr\big(|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star| > \varepsilon\big) \leq \Big(2^{\beta'} C_B + (2R)^{\beta'} C_S\Big) n^{-\alpha'} \varepsilon^{-\beta'}.$$

Therefore the claim holds with

$$C_{\mathrm{PATSO}} := 2^{\beta'} C_B + (2R)^{\beta'} C_S.$$

This completes the proof. □

Theorem D.11 (Polynomial tail currently supported by PATSO)**.** *Assume Assumption D.7. Assume that the Lemma D.9 holds: for every suboptimal arm $a \neq a_\star$,*

$$\Pr(T_a(n) > u) \leq \frac{\bar{c}_{\mathrm{TS}}}{b-1}(u-1)^{-(b-1)}, \qquad \forall u \geq A_a(n),$$

where

$$A_a(n) := \left\lceil \left(\frac{2C}{\Delta_a}\right)^{\beta/\alpha} n^{b/\alpha} \right\rceil, \qquad \Delta_a := \mu_\star - \mu_a.$$

Define

$$c_\Delta := \max_{a \neq a_\star} \left(\frac{2C}{\Delta_a}\right)^{\beta/\alpha}.$$

Let

$$\widehat{\mu}_n^{\mathrm{PATSO}}(p) := \left(\sum_{a=1}^{K} \frac{T_a(n)}{n} \left[\widehat{\mu}_{a,T_a(n)}\right]^p\right)^{1/p}.$$

Then there exist constants

$$n_0,\ c_0,\ c_1,\ C_1,\ C_2 > 0,$$

depending only on

$$(K, p, \alpha, \beta, b, R, c_\mu, \bar{c}_{\mathrm{TS}}, c_\Delta),$$

such that for every $n \geq n_0$ and every

$$0 < \varepsilon \leq R, \qquad \varepsilon > \max\Big\{c_0\, n^{-(1-b/\alpha)/p},\ c_1\, n^{-\alpha/\beta}\Big\},$$

we have

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\big| > \varepsilon\big) \leq C_1\, n^{-\alpha}\varepsilon^{-\beta} + C_2\, n^{-(b-1)}\varepsilon^{-p(b-1)}.$$

Consequently, if

$$\bar{\alpha} := \min\{\alpha, b-1\}, \qquad \bar{\beta} := \max\{\beta, p(b-1)\},$$

then there exists a constant $C > 0$ such that

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\big| > \varepsilon\big) \leq C\, n^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}$$

for the same range of (n, ε).

Proof. For brevity write

$$w_{a,n} := \frac{T_a(n)}{n}, \qquad e_{a,n} := \left|\widehat{\mu}_{a,T_a(n)} - \mu_a\right|, \qquad S_n := \sum_{a \neq a_\star} w_{a,n}.$$

I. deterministic decomposition. We first compare $\widehat{\mu}_n^{\mathrm{PATSO}}(p)$ to the power mean built from the limiting arm means:

$$M_n := \left(\sum_{a=1}^{K} w_{a,n}\mu_a^p\right)^{1/p}.$$

By Minkowski's inequality,

$$\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - M_n\right| \leq \left(\sum_{a=1}^{K} w_{a,n} e_{a,n}^p\right)^{1/p}.$$

Since $\mu_a \leq \mu_\star$ for all a and $a_\star$ is unique,

$$M_n \leq \mu_\star,$$

and moreover

$$\mu_\star - M_n \leq \mu_\star - \mu_\star(1 - S_n)^{1/p} \leq \mu_\star S_n \leq R S_n,$$

because $x \mapsto x^{1/p}$ is concave on $[0, 1]$ and $(1 - s)^{1/p} \geq 1 - s$ for $s \in [0, 1]$.

Therefore,

$$\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\right| \le \left(\sum_{a=1}^{K} w_{a,n} e_{a,n}^p\right)^{1/p} + RS_n.$$

Now,

$$\sum_{a=1}^{K} w_{a,n} e_{a,n}^p = w_{a_\star,n} e_{a_\star,n}^p + \sum_{a \ne a_\star} w_{a,n} e_{a,n}^p \le e_{a_\star,n}^p + R^p S_n,$$

since $0 \le e_{a,n} \le R$. Hence

$$\left(\sum_{a=1}^{K} w_{a,n} e_{a,n}^p\right)^{1/p} \le \left(e_{a_\star,n}^p + R^p S_n\right)^{1/p} \le e_{a_\star,n} + RS_n^{1/p}.$$

Because $0 \le S_n \le 1$ and $p \ge 1$, we have $S_n \le S_n^{1/p}$. Thus

$$\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\right| \le e_{a_\star,n} + 2RS_n^{1/p}.$$

Consequently,

$$\Pr\left(\left|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\right| > \varepsilon\right) \le \Pr\left(e_{a_\star,n} > \frac{\varepsilon}{2}\right) + \Pr\left(S_n > \left(\frac{\varepsilon}{4R}\right)^p\right).$$

It remains to control the two terms on the right-hand side.

II. control of the suboptimal-mass term. Set

$$\delta := \left(\frac{\varepsilon}{4R}\right)^p.$$

If $S_n > \delta$, then

$$\sum_{a \ne a_\star} T_a(n) > \delta n.$$

Hence, by the pigeonhole principle, there exists some $a \ne a_\star$ such that

$$T_a(n) > \frac{\delta n}{K-1}.$$

Let

$$u_\delta := \left\lfloor \frac{\delta n}{2(K-1)} \right\rfloor.$$

Then the event $\{S_n > \delta\}$ implies

$$\exists a \ne a_\star : \; T_a(n) > u_\delta.$$

Therefore,

$$\Pr(S_n > \delta) \le \sum_{a \ne a_\star} \Pr(T_a(n) > u_\delta).$$

Now choose

$$c_0 := 4R\,(4(K-1)c_\Delta)^{1/p}.$$

If

$$\varepsilon > c_0\, n^{-(1-b/\alpha)/p},$$

then

$$\frac{\delta n}{2(K-1)} = \frac{1}{2(K-1)} \left(\frac{\varepsilon}{4R}\right)^p n \ge 2c_\Delta n^{b/\alpha},$$

and thus, for all sufficiently large n,

$$u_\delta \ge A_a(n) \qquad \text{for every } a \ne a_\star.$$

Hence, by the assumed strict pull-tail bound,

$$\Pr(S_n > \delta) \le \sum_{a \ne a_\star} \frac{\bar{c}_{\mathrm{TS}}}{b-1} (u_\delta - 1)^{-(b-1)} \le C_2\, n^{-(b-1)} \varepsilon^{-p(b-1)}$$

for some constant $C_2 > 0$.

III. control of the optimal-arm estimation term. Let

$$\eta := \frac{\varepsilon}{2}.$$

We decompose

$$\Pr(e_{a_\star,n} > \eta) \le \Pr\Big(T_{a_\star}(n) < \frac{n}{2}\Big) + \Pr\Big(e_{a_\star,n} > \eta,\ T_{a_\star}(n) \ge \frac{n}{2}\Big).$$

For the first term, if $T_{a_\star}(n) < n/2$, then

$$\sum_{a \ne a_\star} T_a(n) > \frac{n}{2},$$

so again by the pigeonhole principle there exists $a \ne a_\star$ with

$$T_a(n) > \frac{n}{2(K-1)}.$$

Since $b < \alpha$, we have $n^{b/\alpha} = o(n)$, hence for all sufficiently large n,

$$\frac{n}{2(K-1)} \ge A_a(n) \qquad \forall a \ne a_\star.$$

Thus

$$\Pr\Big(T_{a_\star}(n) < \frac{n}{2}\Big) \le \sum_{a \ne a_\star} \Pr\Big(T_a(n) > \frac{n}{2(K-1)}\Big) \le C_3\, n^{-(b-1)}$$

for some constant $C_3 > 0$.

For the second term,

$$\Pr\Big(e_{a_\star,n} > \eta,\ T_{a_\star}(n) \ge \frac{n}{2}\Big) \le \sup_{m \ge n/2} \Pr(|\widehat{\mu}_{a_\star,m} - \mu_\star| > \eta).$$

By Assumption D.7,

$$\Pr(|\widehat{\mu}_{a_\star,m} - \mu_\star| > \eta) \le c_\mu\, m^{-\alpha} \eta^{-\beta},$$

provided $\eta > m^{-\alpha/\beta}$. Choose

$$c_1 := 2^{1+\alpha/\beta}.$$

If

$$\varepsilon > c_1\, n^{-\alpha/\beta},$$

then

$$\eta = \frac{\varepsilon}{2} > \Big(\frac{n}{2}\Big)^{-\alpha/\beta},$$

and therefore for all $m \ge n/2$,

$$\Pr(|\widehat{\mu}_{a_\star,m} - \mu_\star| > \eta) \le c_\mu\, 2^\alpha n^{-\alpha} \eta^{-\beta} \le C_1\, n^{-\alpha} \varepsilon^{-\beta}$$

for some constant $C_1 > 0$.

Combining the two pieces,

$$\Pr(e_{a_\star,n} > \eta) \le C_3 n^{-(b-1)} + C_1 n^{-\alpha} \varepsilon^{-\beta}.$$

IV. combine the bounds. Putting Steps 2 and 3 together,

$$\Pr\big(|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star| > \varepsilon\big) \le C_3 n^{-(b-1)} + C_1 n^{-\alpha} \varepsilon^{-\beta} + C_2 n^{-(b-1)} \varepsilon^{-p(b-1)}.$$

Because $0 < \varepsilon \le R$,

$$n^{-(b-1)} \le R^{p(b-1)} n^{-(b-1)} \varepsilon^{-p(b-1)},$$

so the first term is absorbed into the third, yielding

$$\Pr\big(|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star| > \varepsilon\big) \le C_1\, n^{-\alpha} \varepsilon^{-\beta} + C_2\, n^{-(b-1)} \varepsilon^{-p(b-1)}.$$

For the final one-piece bound, define

$$\bar{\alpha} := \min\{\alpha, b-1\}, \qquad \bar{\beta} := \max\{\beta, p(b-1)\}.$$

Since $0 < \varepsilon \le R$,

$$n^{-\alpha}\varepsilon^{-\beta} \le R^{\bar{\beta}-\beta}\, n^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}},$$

and

$$n^{-(b-1)}\varepsilon^{-p(b-1)} \le R^{\bar{\beta}-p(b-1)}\, n^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}.$$

Hence

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{PATSO}}(p) - \mu_\star\big| > \varepsilon\big) \le C\, n^{-\bar{\alpha}}\varepsilon^{-\bar{\beta}}$$

for a suitable constant $C > 0$. □

Remark D.12. Given Lemma **??** and Lemma D.9, the remainder of the proof is the same deterministic power-mean reduction as in the scalar theorem: one controls the suboptimal-mass term through Lemma D.9 and the within-arm estimation term through Assumption D.7.

Corollary D.13 (CATSO with discretization error)**.** *Suppose the assumptions of Theorem D.11 hold, except that the maintained return law is projected onto a categorical grid with uniform mean bias bounded by η_{cat}. Then there exists a constant $C_{\mathrm{CATSO}} > 0$ such that, for all $n \ge 1$ and all $\varepsilon > n^{-\alpha'/\beta'}$,*

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \mu_\star\big| > \varepsilon + \eta_{\mathrm{cat}}\big) \le C_{\mathrm{CATSO}}\, n^{-\alpha'}\varepsilon^{-\beta'}.$$

In particular, if $\eta_{\mathrm{cat}} \le c_{\mathrm{cat}}\Delta_z$, then

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \mu_\star\big| > \varepsilon + c_{\mathrm{cat}}\Delta_z\big) \le C_{\mathrm{CATSO}}\, n^{-\alpha'}\varepsilon^{-\beta'}.$$

Corollary D.14 (Capped PATSO)**.** *Suppose the assumptions of Theorem D.11 hold and the cap-and-merge mechanism induces a uniform mean bias bounded by $\eta_{\mathrm{cap}}(K)$. Then there exists a constant $C_{\mathrm{cPATSO}} > 0$ such that, for all $n \ge 1$ and all $\varepsilon > n^{-\alpha'/\beta'}$,*

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{cPATSO}}(p) - \mu_\star\big| > \varepsilon + \eta_{\mathrm{cap}}(K)\big) \le C_{\mathrm{cPATSO}}\, n^{-\alpha'}\varepsilon^{-\beta'}.$$

In particular, if $\eta_{\mathrm{cap}}(K) \le c_{\mathrm{cap}}/K$, then

$$\Pr\Big(\big|\widehat{\mu}_n^{\mathrm{cPATSO}}(p) - \mu_\star\big| > \varepsilon + \frac{c_{\mathrm{cap}}}{K}\Big) \le C_{\mathrm{cPATSO}}\, n^{-\alpha'}\varepsilon^{-\beta'}.$$

Assumption D.15 (CATSO arm-level concentration with discretization bias)**.** For each arm $a \in [K]$ and each local sample size $m \ge 1$, let $\mathcal{G}_{a,m}$ be the σ-field generated by the first m observed returns from arm a. Let $\phi_{a,m}^{\mathrm{CATSO}}$ denote the CATSO Thompson draw after m pulls of arm a, and define the maintained mean

$$\widehat{\mu}_{a,m}^{\mathrm{CATSO}} := \mathbb{E}\big[\phi_{a,m}^{\mathrm{CATSO}} \,\big|\, \mathcal{G}_{a,m}\big].$$

Assume that there exist constants $\alpha, \beta, c_\mu > 0$, numbers $\bar{\mu}_a \in [0, R]$, and a discretization level $\eta_{\mathrm{cat}} \ge 0$ such that

$$|\bar{\mu}_a - \mu_a| \le \eta_{\mathrm{cat}} \qquad \forall a \in [K],$$

and for all $a \in [K]$, all $m \ge 1$, and all $\varepsilon > m^{-\alpha/\beta}$,

$$\Pr\big(\big|\widehat{\mu}_{a,m}^{\mathrm{CATSO}} - \bar{\mu}_a\big| > \varepsilon\big) \le c_\mu\, m^{-\alpha}\varepsilon^{-\beta}.$$

Assumption D.16 (CATSO conditional index tails)**.** There exists $c_{\mathrm{idx}} > 0$ such that for every round $t \ge K$, every arm $c \in [K]$, every $m \in \{1, \dots, t\}$, and every history $\mathcal{F}_t$ with $T_c(t) = m$,

$$\Pr\left(\phi_{c,m}^{(t),\mathrm{CATSO}} - \bar{\mu}_c > C\,\frac{t^{b/\beta}}{m^{\alpha/\beta}} \,\middle|\, \mathcal{F}_t\right) \le c_{\mathrm{idx}}\, C^{-\beta} t^{-b},$$

and

$$\Pr\left(\bar{\mu}_c - \phi_{c,m}^{(t),\mathrm{CATSO}} > C\,\frac{t^{b/\beta}}{m^{\alpha/\beta}} \,\middle|\, \mathcal{F}_t\right) \le c_{\mathrm{idx}}\, C^{-\beta} t^{-b}.$$

Assumption D.17 (CATSO weighted random-time bound)**.** Let

$$\alpha' := (b-1)\left(1-\frac{b}{\alpha}\right), \qquad \beta' := b-1.$$

Assume there exist constants $C_B > 0$ and $c_B > 0$ such that for all $n \geq 1$ and all

$$x \geq c_B\, n^{-p(1-b/\alpha)},$$

$$\Pr\left(\sum_{a=1}^{K} \frac{T_a(n)}{n} \left|\widehat{\mu}^{\mathrm{CATSO}}_{a,T_a(n)} - \bar{\mu}_a\right|^p > x\right) \leq C_B\, n^{-\alpha'} x^{-\beta'/p}.$$

Lemma D.18 (Suboptimal pull tail for CATSO under conditional index tails)**.** *Assume Assumption D.16. Let*

$$\bar{\mu}_\star := \max_{a\in[K]} \bar{\mu}_a, \qquad \bar{a}_\star \in \arg\max_{a\in[K]} \bar{\mu}_a,$$

and assume $\bar{a}_\star$ is unique. For $a \neq \bar{a}_\star$, define

$$\bar{\Delta}_a := \bar{\mu}_\star - \bar{\mu}_a, \qquad A_a^{\mathrm{CATSO}}(n) := \left\lceil \left(\frac{2C}{\bar{\Delta}_a}\right)^{\beta/\alpha} n^{b/\alpha} \right\rceil.$$

If each arm is sampled once initially and for all $t \geq K$,

$$A_{t+1} \in \arg\max_{c\in[K]} \left\{ \phi^{(t),\mathrm{CATSO}}_{c,T_c(t)} + C\, \frac{t^{b/\beta}}{T_c(t)^{\alpha/\beta}} \right\},$$

then for every suboptimal arm $a \neq \bar{a}_\star$ and every integer $u \geq A_a^{\mathrm{CATSO}}(n)$,

$$\Pr(T_a(n) > u) \leq \frac{2c_{\mathrm{idx}}\, C^{-\beta}}{b-1}\, (u-1)^{-(b-1)}.$$

Proof. The proof is verbatim the proof of Lemma D.9 after the substitutions

$$\mu_c \mapsto \bar{\mu}_c, \qquad \mu_\star \mapsto \bar{\mu}_\star, \qquad \Delta_a \mapsto \bar{\Delta}_a, \qquad \phi^{(t),\mathrm{PATSO}}_{c,m} \mapsto \phi^{(t),\mathrm{CATSO}}_{c,m}.$$

□

Lemma D.19 (Suboptimal-mass tail for CATSO)**.** *Let*

$$\alpha' := (b-1)\left(1-\frac{b}{\alpha}\right), \qquad \beta' := b-1,$$

and define

$$S_n^{\mathrm{CATSO}} := \sum_{a\neq\bar{a}_\star} \frac{T_a(n)}{n}.$$

Assume Lemma D.18. Let

$$c_\Delta^{\mathrm{CATSO}} := \max_{a\neq\bar{a}_\star} \left(\frac{2C}{\bar{\Delta}_a}\right)^{\beta/\alpha},$$

and

$$c_S^{\mathrm{CATSO}} := 4(K-1)\big(c_\Delta^{\mathrm{CATSO}} + 1\big).$$

Then there exists a constant $C_S^{\mathrm{CATSO}} > 0$ such that, for all $n \geq 1$ and all

$$y \geq c_S^{\mathrm{CATSO}}\, n^{-(1-b/\alpha)},$$

we have

$$\Pr\big(S_n^{\mathrm{CATSO}} > y\big) \leq C_S^{\mathrm{CATSO}}\, n^{-\alpha'} y^{-\beta'}.$$

One may take

$$C_S^{\mathrm{CATSO}} := \frac{2^{\beta'}(K-1)^{\beta'+1}\, 2c_{\mathrm{idx}}\, C^{-\beta}}{b-1}.$$

Proof. Fix $y > 0$ and define

$$u_y := \left\lfloor \frac{yn}{K-1} \right\rfloor.$$

If $S_n^{\mathrm{CATSO}} > y$, then

$$\sum_{a \neq \bar{a}_\star} T_a(n) > yn,$$

so by the pigeonhole principle there exists some $a \neq \bar{a}_\star$ such that

$$T_a(n) > \frac{yn}{K-1} \geq u_y.$$

Hence

$$\{S_n^{\mathrm{CATSO}} > y\} \subseteq \bigcup_{a \neq \bar{a}_\star} \{T_a(n) > u_y\}.$$

Now assume

$$y \geq c_S^{\mathrm{CATSO}}\, n^{-(1-b/\alpha)}.$$

Then

$$\frac{yn}{K-1} \geq 4\big(c_\Delta^{\mathrm{CATSO}} + 1\big) n^{b/\alpha} \geq 4,$$

which implies

$$u_y - 1 \geq \frac{1}{2} \cdot \frac{yn}{K-1}.$$

Also,

$$u_y \geq \frac{yn}{K-1} - 1 \geq 4\big(c_\Delta^{\mathrm{CATSO}} + 1\big) n^{b/\alpha} - 1 \geq c_\Delta^{\mathrm{CATSO}} n^{b/\alpha} + 1 \geq A_a^{\mathrm{CATSO}}(n)$$

for every $a \neq \bar{a}_\star$.

Therefore, by Lemma D.18,

$$\begin{aligned}
\Pr\big(S_n^{\mathrm{CATSO}} > y\big) &\leq \sum_{a \neq \bar{a}_\star} \Pr(T_a(n) > u_y) \\
&\leq \frac{(K-1)\, 2c_{\mathrm{idx}}\, C^{-\beta}}{b-1} (u_y - 1)^{-\beta'} \\
&\leq \frac{2^{\beta'} (K-1)^{\beta'+1}\, 2c_{\mathrm{idx}}\, C^{-\beta}}{b-1}\, n^{-\beta'} y^{-\beta'}.
\end{aligned}$$

Since

$$\alpha' = \beta' \left(1 - \frac{b}{\alpha}\right) \leq \beta',$$

we have $n^{-\beta'} \leq n^{-\alpha'}$ for all $n \geq 1$, and thus

$$\Pr\big(S_n^{\mathrm{CATSO}} > y\big) \leq C_S^{\mathrm{CATSO}}\, n^{-\alpha'} y^{-\beta'}.$$

This completes the proof. □

Theorem D.20 (CATSO concentrates polynomially around its discretized target)**.** *Let*

$$\alpha' := (b-1)\left(1 - \frac{b}{\alpha}\right), \qquad \beta' := b - 1.$$

Assume Assumptions D.15, D.16, D.17, and Lemma D.18. Define

$$\widehat{\mu}_n^{\mathrm{CATSO}}(p) := \left(\sum_{a=1}^{K} \frac{T_a(n)}{n} \left[\widehat{\mu}_{a,T_a(n)}^{\mathrm{CATSO}}\right]^p \right)^{1/p}.$$

Then there exist constants $c_\varepsilon > 0$ and $C_{\mathrm{CATSO}} > 0$ such that for all $n \geq 1$ and all

$$\varepsilon \geq c_\varepsilon\, n^{-(1-b/\alpha)},$$

we have

$$\Pr\big(|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star| > \varepsilon\big) \le C_{\mathrm{CATSO}}\, n^{-\alpha'} \varepsilon^{-\beta'}.$$

In particular, after enlarging the constant if necessary,

$$\widehat{\mu}_n^{\mathrm{CATSO}}(p) \xrightarrow[\alpha',\beta']{n\to\infty} \bar{\mu}_\star$$

in the sense of Definition D.6.

Proof. For brevity write

$$w_{a,n} := \frac{T_a(n)}{n}, \qquad S_n := S_n^{\mathrm{CATSO}} = \sum_{a\neq \bar{a}_\star} w_{a,n},$$

and define

$$B_n := \left(\sum_{a=1}^{K} w_{a,n}\, |\widehat{\mu}_{a,T_a(n)}^{\mathrm{CATSO}} - \bar{\mu}_a|^p\right)^{1/p}.$$

Also define the comparison power mean

$$M_n := \left(\sum_{a=1}^{K} w_{a,n}\, \bar{\mu}_a^p\right)^{1/p}.$$

By the reverse triangle inequality in ℓ_p,

$$|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - M_n| \le B_n.$$

Next, since $\bar{\mu}_a \le \bar{\mu}_\star$ for all a,

$$M_n \le \bar{\mu}_\star.$$

Moreover,

$$M_n \ge (w_{\bar{a}_\star,n}\, \bar{\mu}_\star^p)^{1/p} = \bar{\mu}_\star\, w_{\bar{a}_\star,n}^{1/p} = \bar{\mu}_\star (1 - S_n)^{1/p}.$$

Hence

$$0 \le \bar{\mu}_\star - M_n \le \bar{\mu}_\star\Big(1 - (1 - S_n)^{1/p}\Big).$$

Since $p \ge 1$,

$$(1-s)^{1/p} \ge 1 - s \qquad \forall s \in [0,1],$$

so

$$0 \le \bar{\mu}_\star - M_n \le \bar{\mu}_\star S_n \le R S_n.$$

Combining the two displays gives

$$|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star| \le B_n + R S_n.$$

Therefore, for every $\varepsilon > 0$,

$$\Pr\big(|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star| > \varepsilon\big) \le \Pr\Big(B_n > \frac{\varepsilon}{2}\Big) + \Pr\Big(S_n > \frac{\varepsilon}{2R}\Big). \tag{84}$$

We first bound the B_n term. If

$$\varepsilon \ge 2 c_B^{1/p}\, n^{-(1-b/\alpha)},$$

then

$$\left(\frac{\varepsilon}{2}\right)^p \ge c_B\, n^{-p(1-b/\alpha)},$$

and Assumption D.17 yields

$$\begin{aligned}
\Pr\Big(B_n > \frac{\varepsilon}{2}\Big) &= \Pr\Big(B_n^p > \left(\frac{\varepsilon}{2}\right)^p\Big) \\
&\le C_B\, n^{-\alpha'} \left(\left(\frac{\varepsilon}{2}\right)^p\right)^{-\beta'/p} \\
&= 2^{\beta'} C_B\, n^{-\alpha'} \varepsilon^{-\beta'}.
\end{aligned}$$

Next, by Lemma D.19, if

$$\varepsilon \geq 2R\, c_S^{\mathrm{CATSO}}\, n^{-(1-b/\alpha)},$$

then

$$\frac{\varepsilon}{2R} \geq c_S^{\mathrm{CATSO}}\, n^{-(1-b/\alpha)},$$

and therefore

$$\begin{aligned}\Pr\Big(S_n > \frac{\varepsilon}{2R}\Big) &\leq C_S^{\mathrm{CATSO}}\, n^{-\alpha'} \left(\frac{\varepsilon}{2R}\right)^{-\beta'} \\ &= (2R)^{\beta'} C_S^{\mathrm{CATSO}}\, n^{-\alpha'} \varepsilon^{-\beta'}.\end{aligned}$$

Now choose

$$c_\varepsilon := \max\Big\{2c_B^{1/p},\ 2R\, c_S^{\mathrm{CATSO}}\Big\},$$

and

$$C_{\mathrm{CATSO}} := 2^{\beta'} C_B + (2R)^{\beta'} C_S^{\mathrm{CATSO}}.$$

Then for every

$$\varepsilon \geq c_\varepsilon\, n^{-(1-b/\alpha)},$$

both bounds apply, and (84) gives

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star\big| > \varepsilon\big) \leq C_{\mathrm{CATSO}}\, n^{-\alpha'} \varepsilon^{-\beta'}.$$

Finally, set

$$\widetilde{C}_{\mathrm{CATSO}} := \max\{C_{\mathrm{CATSO}},\, c_\varepsilon^{\beta'},\, 1\}.$$

If

$$n^{-\alpha'/\beta'} < \varepsilon < c_\varepsilon\, n^{-\alpha'/\beta'},$$

then

$$n^{-\alpha'} \varepsilon^{-\beta'} > c_\varepsilon^{-\beta'},$$

hence

$$1 \leq \widetilde{C}_{\mathrm{CATSO}}\, n^{-\alpha'} \varepsilon^{-\beta'}.$$

Since probabilities are at most 1, the desired bound also holds in this regime. □

Corollary D.21 (CATSO concentration around the true optimum up to discretization error)**.** *Under the assumptions of Theorem D.20, for all $n \geq 1$ and all*

$$\varepsilon \geq c_\varepsilon\, n^{-(1-b/\alpha)},$$

we have

$$\Pr\big(\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \mu_\star\big| > \varepsilon + \eta_{\mathrm{cat}}\big) \leq C_{\mathrm{CATSO}}\, n^{-\alpha'} \varepsilon^{-\beta'}.$$

Proof. Since

$$|\bar{\mu}_a - \mu_a| \leq \eta_{\mathrm{cat}} \qquad \forall a \in [K],$$

taking maxima gives

$$\mu_\star - \eta_{\mathrm{cat}} \leq \bar{\mu}_\star \leq \mu_\star + \eta_{\mathrm{cat}},$$

hence

$$|\bar{\mu}_\star - \mu_\star| \leq \eta_{\mathrm{cat}}.$$

Therefore

$$\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \mu_\star\big| \leq \big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star\big| + |\bar{\mu}_\star - \mu_\star| \leq \big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star\big| + \eta_{\mathrm{cat}}.$$

Hence

$$\big\{\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \mu_\star\big| > \varepsilon + \eta_{\mathrm{cat}}\big\} \subseteq \big\{\big|\widehat{\mu}_n^{\mathrm{CATSO}}(p) - \bar{\mu}_\star\big| > \varepsilon\big\}.$$

Applying Theorem D.20 completes the proof. □

Corollary D.22 (Uniform-grid CATSO)**.** *Assume the assumptions of Theorem D.20. If CATSO projects each backed-up return onto its nearest atom on a uniform grid with maximal spacing Δ_z, then*

$$\eta_{\text{cat}} \leq \frac{\Delta_z}{2}.$$

Consequently, for all $n \geq 1$ and all

$$\varepsilon \geq c_\varepsilon \, n^{-(1-b/\alpha)},$$

$$\Pr\left(\left| \widehat{\mu}_n^{\text{CATSO}}(p) - \mu_\star \right| > \varepsilon + \frac{\Delta_z}{2} \right) \leq C_{\text{CATSO}} \, n^{-\alpha'} \varepsilon^{-\beta'}.$$

Proof. Nearest-atom projection on a uniform grid of spacing Δ_z incurs pointwise error at most $\Delta_z/2$. Therefore the corresponding mean bias is also at most $\Delta_z/2$, so

$$|\bar{\mu}_a - \mu_a| \leq \frac{\Delta_z}{2} \qquad \forall a \in [K].$$

Apply Corollary D.21. □

E Convergence of CATSO and PATSO in Monte-Carlo Tree Search

Based upon the results of CATSO and PATSO using power mean as the value backup operator on the described non-stationary multi-armed bandit problem, we derive theoretical results for CATSO in an MCTS tree.

We derive Theorem E for CATSO and Theorem E for PATSO, which show concentration and convergence for any internal node in the tree. These proofs utilize induction, leveraging the results of Lemma D.3 for CATSO and Lemma D.4 for PATSO, and Lemma D.1 for CATSO and Lemma D.2 for PATSO. Additionally, we use Lemma C.1, which demonstrates the concentration and convergence of an estimated Q-value based on the child V-value node, applying it recursively throughout the tree.

Our main results, Theorem E for CATSO and Theorem E for PATSO, show that the simple regret converges non-asymptotically at a rate of $O(n^{-1/2})$.

Theorem 3. *Let CATSO run on a tree of depth H. For each node s_h at level h, the following statements hold:*

i) For any action $a_k \in \mathcal{A}_{s_h}$,

$$\underset{n\to\infty}{\text{plim}} \, \widehat{Q}_n(s_h, a_k) = \widetilde{Q}(s_h, a_k).$$

ii)

$$\underset{n\to\infty}{\text{plim}} \, \widehat{V}_n(s_h) = \widetilde{V}(s_h).$$

Proof. We prove this by induction on the maximum depth D of the search tree.

Base Case ($D = 1$). Consider a tree of depth 1 whose root node is s_0. When taking action a_k from s_0, the algorithm observes an immediate reward $r_t(s_0, a_k)$ and transitions to a leaf s_1. Let

$$R(s_0, a_k) \;=\; \mathbb{E}\big[r_t(s_0, a_k)\big]$$

denote the mean immediate reward, and recall

$$\widetilde{Q}(s_0, a_k) \;=\; R(s_0, a_k) \;+\; \gamma \sum_{s_1 \in \mathcal{A}_{s_0}} \mathbb{P}(s_1 \mid s_0, a_k) \, \widetilde{V}(s_1).$$

We can see CATSO estimate Q value (this is straight forward as simple derivation from psudocode) as

$$\widehat{Q}_n(s_0, a_k) \;=\; \frac{1}{n} \sum_{t=1}^{n} r_t(s_0, a_k) \;+\; \gamma \sum_{s_1 \sim \tau(s_0, a_k)} \frac{T_{s_0, a_k}^{s_1}(n)}{n} \, \widehat{V}_{T_{s_0, a_k}^{s_1}(n)}(s_1),$$

where $T_{s_0, a_k}^{s_1}(n)$ is the number of visits at s_0, a_k, s_1

First, by Lemma C.1 (treating $\{r_t\}$ as i.i.d. intermediate rewards with child-state probabilities p_m), one obtains

$$\operatorname*{plim}_{n\to\infty} \widehat{Q}_n(s_0, a_k) = \widetilde{Q}(s_0, a_k).$$

Second, from Theorem **??** (or directly from plugging in the limit above), we deduce

$$\operatorname*{plim}_{n\to\infty} \widehat{V}_n(s_0) = \widetilde{V}(s_0).$$

since $\widetilde{V}(s_0) = \max_{a\in\mathcal{A}_{s_0}} \widetilde{Q}(s_0, a)$ and

$$\widehat{V}_n(s_0) \;=\; \Big(\sum_{a\in\mathcal{A}_{s_0}} \frac{T_{s_0,a}(n)}{n} \Big[\widehat{Q}_{T_{s_0,a}(n)}(s_0, a)\Big]^p \Big)^{\frac{1}{p}}.$$

Hence, the result holds for $D = 1$.

Inductive Step. Suppose the statements (i) and (ii) are valid for all nodes in any tree of depth $\leq D$. We now verify that they also hold for a tree of depth $D + 1$.

At the root s_0, after taking action a_k, the algorithm transitions to a subtree of depth D. By the inductive hypothesis, for every node in that subtree (including its root $s_1 \sim \tau(s_0, a_k)$), the limit properties of Q and V hold. Hence, each $\widehat{V}_n(s_1)$ converges to its corresponding $\widetilde{V}(s_1)$, ensuring that

$$\operatorname*{plim}_{n\to\infty} \widehat{Q}_n(s_0, a_k) \;=\; R(s_0, a_k) \;+\; \gamma \sum_{s_1\sim\tau(s_0,a_k)} \mathbb{P}(s_1 \mid s_0, a_k) \text{ and } \operatorname*{plim}_{n\to\infty} \widehat{V}_n(s_1) \;=\; \widetilde{Q}(s_0, a_k).$$

To conclude (ii), note that

$$\widetilde{V}(s_0) = \max_{a\in\mathcal{A}_{s_0}} \widetilde{Q}(s_0, a), \quad \text{while} \quad \widehat{V}_n(s_0) = \Big(\sum_{a\in\mathcal{A}_{s_0}} \frac{T_{s_0,a}(n)}{n} \Big[\widehat{Q}_{T_{s_0,a}(n)}(s_0, a)\Big]^p \Big)^{\frac{1}{p}}.$$

Since each $\widehat{Q}_n(s_0, a)$ converges to the correct $\widetilde{Q}(s_0, a)$, it follows immediately that

$$\operatorname*{plim}_{n\to\infty} \widehat{V}_n(s_0) \;=\; \widetilde{V}(s_0).$$

Therefore, by induction, both statements (i) and (ii) hold for every node in a tree of any finite depth H. This completes the proof. □

Similarly we can derive the following Theorem

Theorem 4.

Consider a tree of maximum depth H on which `PATSO` *is run. Then, for any node s_h at depth h and any action $a_k \in \mathcal{A}_{s_h}$:*

i)

$$\operatorname*{plim}_{n\to\infty} \widehat{Q}_n(s_h, a_k) = \widetilde{Q}(s_h, a_k).$$

ii)

$$\operatorname*{plim}_{n\to\infty} \widehat{V}_n(s_h) = \widetilde{V}(s_h).$$

Proof. The proof follows the same steps as Theorem E by applying the results of Lemma C.1 and Theorem **??**. □

Notation. At a node s, $T_s(n) = \sum_a T_{s,a}(n)$ is the visit count and the p–power–mean backup is

$$\widehat{V}_n(s) = \left(\sum_a \frac{T_{s,a}(n)}{T_s(n)} \left(\widehat{Q}_n(s,a)\right)^p \right)^{1/p}, \qquad p \geq 1. \tag{85}$$

Let $\mu_{a,n}(s)$ denote the *drifting bandit mean* at (s,a) and $\mu_n^\star(s) = \max_a \mu_{a,n}(s)$. Define the *drift* $\delta_n^\star(s) := \widetilde{V}(s) - \mu_n^\star(s) \geq 0$.

Theorem 5.*(Convergence of Expected Payoff of `CATSO`)* *We have at the root node s_0, there exists a constant $C' > 0$ such that*

$$\mathbb{E}\big[\big|\widehat{V}_n(s_0) - \widetilde{V}(s_0)\big|\big] \;\leq\; C'\, n^{-\frac{1}{2}},$$

Proof. We argue in four steps.

1 (One–step decomposition at a node). For any node s,

$$\mathbb{E}\big|\widehat{V}_n(s) - \widetilde{V}(s)\big| \;\leq\; \underbrace{\mathbb{E}\big|\widehat{V}_n(s) - \mu_n^\star(s)\big|}_{\text{statistical term}} \;+\; \underbrace{\delta_n^\star(s)}_{\text{drift term}}. \tag{86}$$

2 (Control of the statistical term via Lemma 7 at the node). Lemma 7 (Appendix C) gives, for CATSO and any $p \geq 1$,

$$\mathbb{E}\Big|\widehat{V}_n(s) - \mu_n^\star(s)\Big| \;\leq\; \frac{R}{T_s(n)} \sum_{a \neq a^\star} \mathbb{E}\big[T_{s,a}(n)\big] \;+\; \delta_n^\star(s), \tag{87}$$

where $a^\star \in \arg\max_a \mu_{a,n}(s)$ and rewards are in $[0,R]$. Combining (86) and (87) and bounding the (duplicated) $\delta_n^\star(s)$ only once, we obtain

$$\mathbb{E}\Big|\widehat{V}_n(s) - \widetilde{V}(s)\Big| \;\leq\; \frac{R}{T_s(n)} \sum_{a \neq a^\star} \mathbb{E}\big[T_{s,a}(n)\big] \;+\; \delta_n^\star(s).$$

By Lemma 5 (Appendix C), for any suboptimal arm $a \neq a^\star$,

$$\mathbb{E}[T_{s,a}(n)] \;\leq\; \frac{(1+\varepsilon_0)\log T_s(n)}{K_{\inf}^{(N)}(F_a, \mu^\star)} \;+\; O\big(\sqrt{T_s(n)}\big).$$

At the root s_0 we have $T_{s_0}(n) = n$, hence

$$\mathbb{E}\Big|\widehat{V}_n(s_0) - \mu_n^\star(s_0)\Big| \;\leq\; C_1 \frac{\log n}{n} \;+\; C_2\, n^{-1/2} \;=\; O\big(n^{-1/2}\big). \tag{88}$$

3 (Drift contraction down the tree via Lemma 1). Pick an optimal root action $a^\star \in \arg\max_a Q^e(s_0,a)$. Since $\mu_n^\star(s_0) \geq \mu_{a^\star,n}(s_0)$,

$$0 \leq \delta_n^\star(s_0) = \widetilde{V}(s_0) - \mu_n^\star(s_0) \leq \widetilde{V}(s_0) - \mu_{a^\star,n}(s_0).$$

By Lemma 1 (Appendix B),

$$\mu_{a^\star,n}(s_0) \;=\; r(s_0,a^\star) + \gamma \sum_{s_1} P(s_1|s_0,a^\star)\, \mathbb{E}\Big[\widehat{V}_n(s_1)\Big],$$

whence

$$\delta_n^\star(s_0) \;\leq\; \gamma \sum_{s_1} P(s_1|s_0,a^\star)\, \mathbb{E}\big|\widehat{V}_n(s_1) - \widetilde{V}(s_1)\big|. \tag{89}$$

Under Lemma 5, suboptimal root actions get only $O(\sqrt{n}) + O(\log n)$ pulls, so $T_{s_0,a^\star}(n) = n - O(\sqrt{n})$ in expectation. Thus each child s_1 on $a^\star$'s branch is visited $\Theta(n)$ times in expectation, and

the nodewise bound (88) applies at s_1 as well, giving $\mathbb{E}|\widehat{V}_n(s_1) - \mu_n^\star(s_1)| = O(n^{-1/2})$. Iterating (89) down the tree (base case at leaves has zero drift by definition),

$$\delta_n^\star(s_0) \;\le\; \gamma C n^{-1/2} + \gamma^2 C n^{-1/2} + \cdots \;\le\; \frac{C}{1-\gamma}\, n^{-1/2},$$

or $CH\, n^{-1/2}$ for finite horizon H if $\gamma = 1$.

4 (Combine). From (88) and the drift bound,

$$\mathbb{E}\big|\widehat{V}_n(s_0) - \widetilde{V}(s_0)\big| \;\le\; O\big(n^{-1/2}\big) + O\big(n^{-1/2}\big) \;=\; O\big(n^{-1/2}\big).$$

□

Alternative Proof of Theorem 5 - Direct Bias Propagation. We prove that $\mathbb{E}[|\hat{V}_n(s_0) - \tilde{V}(s_0)|] \le C' n^{-1/2}$ by directly analyzing the bias at each node level.

1: Base case - Leaf nodes. At leaf nodes (depth H), we have $\hat{V}_n(s_H) = V_0(s_H)$ by definition, so there's no estimation error.

2: Inductive step - Internal nodes. Consider any internal node s_h at depth $h < H$. By Theorem 3, we know:

$$\text{plim}_{n\to\infty} \hat{Q}_n(s_h, a) = \tilde{Q}(s_h, a) \quad \forall a \in \mathcal{A}_{s_h}$$

From Lemma 7 (adapted to the MCTS setting), for each Q-node:

$$\left|\mathbb{E}[\hat{Q}_n(s_h, a)] - \tilde{Q}(s_h, a)\right| \le \frac{C_1}{n^{1/2}}$$

3: Power-mean is 1-Lipschitz. The power-mean operator $\mathcal{P}_p$ defined by

$$\mathcal{P}_p(q_1, \ldots, q_K; w_1, \ldots, w_K) = \left(\sum_{i=1}^{K} w_i q_i^p\right)^{1/p}$$

is 1-Lipschitz with respect to the ℓ_∞ norm on the q_i's when the weights w_i form a probability distribution. That is:

$$|\mathcal{P}_p(q) - \mathcal{P}_p(q')| \le \|q - q'\|_\infty$$

4: Bias propagation. At node s_h, the V-value estimate is:

$$\hat{V}_n(s_h) = \mathcal{P}_p\left(\hat{Q}_{T_{s_h,a_1}(n)}(s_h, a_1), \ldots, \hat{Q}_{T_{s_h,a_K}(n)}(s_h, a_K); \frac{T_{s_h,a_1}(n)}{n}, \ldots, \frac{T_{s_h,a_K}(n)}{n}\right)$$

By the 1-Lipschitz property and the bias bound on each Q-value:

$$\left|\mathbb{E}[\hat{V}_n(s_h)] - \tilde{V}(s_h)\right| \le \left|\mathbb{E}[\hat{V}_n(s_h)] - \mathcal{P}_p(\tilde{Q}(s_h, a_1), \ldots, \tilde{Q}(s_h, a_K))\right| \tag{90}$$

$$\le \max_{a \in \mathcal{A}_{s_h}} \left|\mathbb{E}[\hat{Q}_{T_{s_h,a}(n)}(s_h, a)] - \tilde{Q}(s_h, a)\right| \tag{91}$$

$$\le \frac{C_1}{n^{1/2}} \tag{92}$$

5: Root node conclusion. Since the same $O(n^{-1/2})$ bias bound holds at every node level, and there are at most H levels, we get at the root:

$$\left|\mathbb{E}[\hat{V}_n(s_0)] - \tilde{V}(s_0)\right| \le \frac{C_1}{n^{1/2}}$$

By Jensen's inequality:

$$\mathbb{E}[|\hat{V}_n(s_0) - \tilde{V}(s_0)|] \le \mathbb{E}[|\hat{V}_n(s_0) - \mathbb{E}[\hat{V}_n(s_0)]|] + |\mathbb{E}[\hat{V}_n(s_0)] - \tilde{V}(s_0)|$$

The first term is $O(n^{-1/2})$ by concentration (Theorem 3), and the second term is $O(n^{-1/2})$ by the above analysis. Therefore:

$$\mathbb{E}[|\hat{V}_n(s_0) - \tilde{V}(s_0)|] \le C' n^{-1/2}$$

□

Corollary E.1 (PATSO)**.** *Replacing Lemma 7 by Lemma 8 (Appendix C) in Step 2 yields the same rate for PATSO: there exists $C' > 0$ such that $\mathbb{E}|\widehat{V}_n(s_0) - \widetilde{V}(s_0)| \leq C' n^{-1/2}$.*
Remark E.2 (Why this avoids the tail–bound pitfall). The argument works purely in expectation at each node by combining: (i) the visit–weighted bandit bounds (Lemmas 7/8) and visit counts (Lemmas 5/6), and (ii) the one–step coupling (Lemma 1). No global extension of a polynomial tail beyond its stated range is needed.

Next, we present the results of Theorem E. The proof follows the same steps as Theorem E.

Theorem 6.*(Convergence of Expected Payoff of `PATSO`)* *We have at the root node s_0, there exists a constant $C' > 0$ such that*

$$\mathbb{E}\big[\big|\widehat{V}_n(s_0) - \widetilde{V}(s_0)\big|\big] \;\leq\; C'\, n^{-\frac{1}{2}},$$

Proof. The proof follows the same analysis as in Theorem E. □

F Distributional MCTS and Wasserstein Robust Optimization

The distributional MCTS approach presented in our paper extends naturally beyond sequential decision-making to the broader domain of robust planning under uncertainty. In this section, we establish formal connections between our algorithms (`CATSO` and `PATSO`) and the field of Wasserstein Distributionally Robust Optimization (WDRO), which offers a principled framework for decision-making under distributional uncertainty.

In our paper, both `CATSO` and `PATSO` maintain distributional representations of Q-values:

- `CATSO`: Uses a categorical distribution with atoms at locations $\{z_i(s,a)\}_{i=0}^{N-1}$ and probabilities $\{p_i(s,a)\}_{i=0}^{N-1}$
- `PATSO`: Uses a particle-based representation with particles $\mathcal{S}(s,a)$ and weights $\alpha(s,a)$

These representations capture empirical distributions $\hat{P}_{s,a}$ of returns for each state-action pair (s,a). We now show that our approach can be reinterpreted within the Wasserstein Distributionally Robust MDP (WDRMDP) framework.

The standard WDRMDP formulation is:

$$\max_{\pi} \min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^\pi] \tag{93}$$

where:

- π is the policy being optimized
- $\hat{P}$ is the empirical distribution (from samples)
- $\mathcal{B}_\varepsilon(\hat{P}) = \{P : W_p(P, \hat{P}) \leq \varepsilon\}$ is the Wasserstein ball of radius ε around $\hat{P}$
- W_p is the p-Wasserstein distance

This connection provides a theoretical foundation for applying our methods to robust planning problems beyond standard MCTS settings.

A key advantage of our distributional approach is that it enables rigorous sample complexity guarantees for robust planning. We formalize this through the concept of (ε, δ)-robust policies.
Definition F.1 ((ε,δ)-Robust Policy)**.** A policy π is (ε,δ)-robust if, with probability at least $1-\delta$, its performance is within ε of the optimal robust policy:

$$\mathbb{P}\left[\min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^\pi] \geq \max_{\pi'} \min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^{\pi'}] - \varepsilon\right] \geq 1-\delta \tag{94}$$

Theorem 7 (Sample Complexity for Distributionally Robust Planning)**.** *Let $\mathcal{M}$ be an MDP with state space $\mathcal{S}$, action space $\mathcal{A}$, bounded rewards in $[0, R_{\max}]$, and discount factor γ. To learn an*

(ε, δ)-robust policy using the distributional representations from CATSO/PATSO combined with a concentration-based Wasserstein radius, the required number of samples is:

$$O\left(\left[\frac{R_{\max}^3}{\varepsilon(1-\gamma)^3}\log\left(\frac{H|\mathcal{S}|^2|\mathcal{A}|}{\delta}\right)\right]^{2H}\right). \tag{95}$$

Proof. We provide a complete proof through several carefully developed steps. The proof applies to both CATSO and PATSO, with an additional consideration for discretization error in CATSO.

1: Wasserstein Concentration Bounds. For distributions with bounded support in $[0, R_{\max}]$, we can apply concentration inequalities for the Wasserstein distance between the empirical distribution $\hat{P}_{s,a}$ (based on $n_{s,a}$ i.i.d. samples) and the true distribution $P_{s,a}$.

For the p-Wasserstein distance $(p \geq 1)$, Fournier and Guillin [2015] provide the following concentration inequality

$$\mathbb{P}[W_p(\hat{P}_{s,a}, P_{s,a}) > t] \leq c_1 \exp(-c_2 n_{s,a} t^2) \tag{96}$$

when $n_{s,a}$ is sufficiently large and t is sufficiently small. Here, c_1 and c_2 are constants that depend on the diameter of the support. For rewards bounded in $[0, R_{\max}]$, the diameter of the support for discounted returns is at most $\frac{R_{\max}}{1-\gamma}$. Simplifying the constants, we get

$$\mathbb{P}\left[W_p(\hat{P}_{s,a}, P_{s,a}) > \sqrt{\frac{cR_{\max}^2 \log(1/\delta)}{(1-\gamma)^2 n_{s,a}}}\right] \leq \delta \tag{97}$$

for an appropriate constant $c > 0$. This means that with probability at least $1-\delta$, the Wasserstein distance between the empirical and true distributions is bounded by the right-hand side.

2: Confidence-Adjusted Wasserstein Radius. We need to ensure that for all state-action pairs (s, a), the true distribution is contained within a Wasserstein ball centered at the empirical distribution with high probability. Using the concentration bound from Step 1 and applying a union bound over all $|\mathcal{S}||\mathcal{A}|$ state-action pairs, we set

$$\varepsilon_{s,a}(n) = \sqrt{\frac{cR_{\max}^2 \log(|\mathcal{S}||\mathcal{A}|/\delta)}{(1-\gamma)^2 n_{s,a}}} \tag{98}$$

With this choice, we have

$$\mathbb{P}\left[\forall (s,a) \in \mathcal{S} \times \mathcal{A} : W_p(P_{s,a}, \hat{P}_{s,a}) \leq \varepsilon_{s,a}(n)\right] \geq 1-\delta \tag{99}$$

This means that with probability at least $1-\delta$, all true distributions are contained within their respective Wasserstein balls.

3: Value Function Error Propagation. We now analyze how errors in the empirical distributions propagate to errors in the value functions. We consider two MDPs with reward distributions P and Q where $W_p(P, Q) \leq \varepsilon$.

Let V_1, V_2 be the value function for policy π under P and Q, respectively. For any state s, the Bellman equation gives

$$V_1(s) = \mathbb{E}_{(r,s')\sim P_{s,\pi(s)}}\left[r + \gamma V_1(s')\right] \tag{100}$$

Similarly, for the empirical MDP,

$$V_2(s) = \mathbb{E}_{(r,s')\sim Q_{s,\pi(s)}}\left[r + \gamma V_2(s')\right]. \tag{101}$$

Let $\Delta(s) = V_1(s) - V_2(s)$. We have

$$\begin{aligned}
\Delta(s) &= \mathbb{E}_{P_{s,\pi(s)}}\left[r + \gamma V_1(s')\right] - \mathbb{E}_{Q_{s,\pi(s)}}\left[r + \gamma V_2(s')\right] && (102)\\
&= \mathbb{E}_{P_{s,\pi(s)}}\left[r + \gamma V_1(s')\right] - \mathbb{E}_{Q_{s,\pi(s)}}\left[r + \gamma V_1(s')\right] && (103)\\
&\quad + \mathbb{E}_{Q_{s,\pi(s)}}\left[\gamma V_1(s')\right] - \mathbb{E}_{Q_{s,\pi(s)}}\left[\gamma V_2(s')\right] && (104)\\
&= \underbrace{\mathbb{E}_{P_{s,\pi(s)}}\left[r + \gamma V_1(s')\right] - \mathbb{E}_{Q_{s,\pi(s)}}\left[r + \gamma V_1(s')\right]}_{\text{Distribution approximation error}} + \gamma \mathbb{E}_{Q_{s,\pi(s)}}\left[V_1(s') - V_2(s')\right]. && (105)
\end{aligned}$$

For the first term, we use the key property of Wasserstein distances: if two distributions P and Q satisfy $W_p(P, Q) \leq \varepsilon$, then for any L-Lipschitz function f,

$$|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq L\varepsilon. \tag{106}$$

The function $f(r, s') = r + \gamma V_1(s')$ is Lipschitz with constant $L = 1 + \gamma \frac{R_{\max}}{1-\gamma} = \frac{1}{1-\gamma}$ (since V_1 is bounded by $\frac{R_{\max}}{1-\gamma}$ and has Lipschitz constant $\frac{1}{1-\gamma}$). Therefore

$$\left|\mathbb{E}_{P_{s,\pi(s)}}\left[r + \gamma V_1(s')\right] - \mathbb{E}_{Q_{s,\pi(s)}}\left[r + \gamma V_1(s')\right]\right| \leq \frac{\varepsilon_{s,\pi(s)}(n)}{1-\gamma}. \tag{107}$$

For the second term, we have

$$\gamma \left|\mathbb{E}_{Q_{s,\pi(s)}}\left[V_1(s') - V_2(s')\right]\right| \leq \gamma \max_{s'} |\Delta(s')|. \tag{108}$$

Combining these results, we obtain

$$|\Delta(s)| \leq \frac{\varepsilon_{s,\pi(s)}(n)}{1-\gamma} + \gamma \max_{s'} |\Delta(s')|. \tag{109}$$

Since this inequality holds for all states, it also holds for the state with maximum error

$$\max_s |\Delta(s)| \leq \frac{\max_{s,a} \varepsilon_{s,a}(n)}{1-\gamma} + \gamma \max_s |\Delta(s)| \tag{110}$$

which implies

$$\max_s |\Delta(s)| \leq \frac{1}{1-\gamma} \cdot \frac{\max_{s,a} \varepsilon_{s,a}(n)}{1-\gamma} = \frac{\max_{s,a} \varepsilon_{s,a}(n)}{(1-\gamma)^2}. \tag{111}$$

In other words, we have

$$\max_s |V_1(s) - V_2(s)| \leq \frac{R_{\max}}{(1-\gamma)^2} \max_{s,a} \varepsilon_{s,a}(n). \tag{112}$$

4: Robust Value Function Error. Now we consider the robust value functions. We define

$$V^{\pi,\text{rob}}(s) = \min_{P \in \mathcal{B}_\varepsilon(P_{\text{true}})} \mathbb{E}_P[V^\pi(s)], \quad \hat{V}^{\pi,\text{rob}}(s) = \min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^\pi(s)].$$

Using the results obtained from Step 2 and Step 3, for all π, we have

$$|V^{\pi,\text{rob}}(s) - \hat{V}^{\pi,\text{rob}}(s)| \leq \frac{3R_{\max}}{(1-\gamma)^2} \max_{s,a} \varepsilon_{s,a}(n) \tag{113}$$

and

$$|V^{\pi,\text{rob}}(s) - \mathbb{E}_{P_{true}}[V^\pi(s)]| \leq \frac{R_{\max}}{(1-\gamma)^2} \max_{s,a} \varepsilon_{s,a}(n).$$

Moreover, for `PATSO` policy (which we will denote by π_0), the results from previous sections dictate

$$\mathbb{E}_{P_{true}}[V^{\pi_0}(s)] \geq \max_{\pi'} \mathbb{E}_{P_{true}}[V^{\pi'}(s)]| - \frac{C'}{\sqrt{n}}.$$

We conclude that

$$\mathbb{E}_{P_{true}}[V^{\pi_0}(s)] \geq \max_{\pi'} \min_{P \in \mathcal{B}_\varepsilon(\hat{P})} \mathbb{E}_P[V^{\pi'}] - \frac{4R_{\max}}{(1-\gamma)^2} \max_{s,a} \varepsilon_{s,a}(n) - \frac{C'}{\sqrt{n}}$$

with probability at least $1 - \delta$.

5: Sample Allocation. We note that the bounds derived from the previous steps depend on $n_{s,a} = T_{s,a}(n)$, the (random) number of times the state-action pairs (s, a) have been visited after n steps.

We first note that for some $C_0 > 0$, we have

$$\mathbb{E}\left[\sum_{a\neq a^*} T_{s,a}(n)\right] \leq C_0(|\mathcal{A}|-1)\sqrt{T_s(n)}$$

which implies

$$\mathbb{P}\left[\sum_{a\neq a^*} T_{s,a}(n) \geq \frac{C_0(|\mathcal{A}|-1)}{\delta}\sqrt{T_s(n)}\right] \leq \delta$$

via Markov's inequality. Thus, if we define the event

$$\mathcal{E}_n = \{T_{s,a^*}(n) \geq \frac{1}{2}T_s(n)\}$$

then $\mathbb{P}[\mathcal{E}_n] \geq 1-\delta$ for

$$T_s(n) \geq \left(\frac{2C_0(|\mathcal{A}|-1)}{\delta}\right)^2.$$

Under the event $\mathcal{E}_n$, we define K as the last time (up until step n) that a^* is the selected action. We note that

$$T_{s,a^*}(K) \geq \frac{1}{2}T_s(n),$$

and that for all a

$$\phi_k(s,a) + C\frac{K^{1/4}}{T_{s,a}(K)^{1/2}} \leq \phi_k(s,a^*) + C\frac{K^{1/4}}{T_{s,a^*}(K)^{1/2}} \leq R_{max} + 2C\frac{T_s(n)^{1/4}}{T_s(n)^{1/2}} = R_{max} + 2C.$$

We deduce that

$$T_{s,a}(n) \geq T_{s,a}(K) \geq \frac{C^2K^{1/2}}{(R_{max}+2C)^2} \geq \frac{C^2}{\sqrt{2}(R_{max}+2C)^2}T_s(n)^{1/2}.$$

In other words, we have for all s,

$$\mathbb{P}\left[\min_a T_{s,a}(n) \geq \frac{C^2}{\sqrt{2}(R_{max}+2C)^2}T_s(n)^{1/2}\right] \geq 1-\delta.$$

Using an induction argument on the depth of the decision tree, we deduce that for any state s_h reachable from depth h, we have

$$\mathbb{P}\left[\min_a T_{s_h,a}(n) \geq \frac{C^2}{\sqrt{2}(R_{max}+2C)^2}n^{1/(2h)}\right] \geq (1-\delta)^h$$

for

$$n \geq \left(\frac{2C_0(|\mathcal{A}|-1)}{\delta}\right)^{2h}.$$

This implies

$$\mathbb{P}\left[\max_{s_h,a}\varepsilon_{s,a}(n) \leq \sqrt{\frac{cR_{\max}^2(R_{max}+2C)^2\log(|\mathcal{S}||\mathcal{A}|/\delta)}{\sqrt{2}C^2(1-\gamma)^2n^{1/h}}}\right] \geq (1-\delta)^h. \tag{114}$$

6: Total Sample Complexity.

For a policy to be (ε_0,δ_0)-robust, we need:

$$\frac{4R_{\max}}{(1-\gamma)^2}\max_{s,a}\varepsilon_{s,a}(n) + \frac{C'}{\sqrt{n}} \leq \varepsilon_0 \quad \text{and} \quad |\mathcal{S}|(1-(1-\delta)^H)+\delta \leq \delta_0.$$

Substituting and solving for n while simplifying constant and dominated terms, we deduce that the sample complexity for the PATSO policy to be (ε_0,δ_0)-robust is of order

$$O\left(\left[\frac{R_{\max}^3}{\varepsilon_0(1-\gamma)^3}\log\left(\frac{H|\mathcal{S}|^2|\mathcal{A}|}{\delta_0}\right)\right]^{2H}\right). \tag{115}$$

6: Additional Discretization Error for CATSO. For CATSO, there is an additional source of error due to the discretization of the return distribution into N atoms. This error arises because:

1. The categorical representation divides the range $[Q_{\min}, Q_{\max}]$ into N equal intervals.
2. Any continuous return value must be approximated by the nearest atom.
3. The maximum approximation error for any single return is half the distance between consecutive atoms.

With N atoms spanning a range of diameter $D = Q_{\max} - Q_{\min}$, the maximum discretization error is:

$$e_{\text{disc}} = \frac{D}{2(N-1)} \approx \frac{D}{2N} \tag{116}$$

For discounted returns bounded in $[0, R_{\max}]$, the effective diameter is:

$$D = \frac{R_{\max}}{1-\gamma} \tag{117}$$

Thus, the discretization error is:

$$e_{\text{disc}} = \frac{R_{\max}}{2N(1-\gamma)} \tag{118}$$

When this error propagates through the Bellman operator, it leads to an additional value function error of:

$$e_{\text{disc,V}} = \frac{e_{\text{disc}}}{1-\gamma} = \frac{R_{\max}}{2N(1-\gamma)^2} \tag{119}$$

To ensure this discretization error is at most $\varepsilon/2$ (allowing the remaining $\varepsilon/2$ for statistical estimation error), we need:

$$\frac{R_{\max}}{2N(1-\gamma)^2} \leq \frac{\varepsilon}{2} \tag{120}$$

Solving for N:

$$N \geq \frac{R_{\max}}{\varepsilon(1-\gamma)^2} \tag{121}$$

Simplifying to big-O notation:

$$N = O\left(\frac{R_{\max}}{\varepsilon(1-\gamma)}\right). \tag{122}$$

Unlike `CATSO`, the `PATSO` algorithm does not suffer from discretization error because:

1. `PATSO` directly stores each distinct observed return as a separate particle.
2. When a new return is observed, it either increments the weight of an existing identical particle or adds a new particle with that exact value.

This representation is mathematically identical to the empirical distribution:

$$W_p(\mathcal{Q}_n^{\texttt{PATSO}}(s,a), \mathcal{Q}_{\text{empirical}}(s,a)) = 0 \quad \forall n \tag{123}$$

Therefore, there is no approximation error in representing the empirical distribution, and `PATSO` only needs to account for the statistical estimation error (i.e., the difference between the empirical and true distributions). The particle-based representation provides an exact representation of the empirical distribution, making it particularly well-suited for capturing complex, multi-modal return distributions without introducing additional approximation errors. □

G Full Experimental Details

All experiments were conducted on 8 Intel Xeon Gold 6130 (Skylake) processors, x86_64 architecture, 2.10 GHz, with 2 CPUs per node and 16 cores per CPU. Whenever feasible, we opted for open-source implementations of algorithms and environments.

G.1 Parameter Selection

We search the number of atoms from $\{10, 20, \ldots, 100\}$ and choose the results with best performance. We set the discount factor $\gamma = 0.99$. For UCT, we use the exploration constant $C = \sqrt{2} \times (R_{\max} - R_{\min})$.

Table 4: Experimental settings overview.

Aspect	Setting
Game list	12 games: Alien, Atlantis, BeamRider, Enduro, Frostbite, Gopher, Hero, MsPacman, Phoenix, Robotank, Seaquest, SpaceInvaders
Random seeds	10 seeds $\times$ method $\times$ game; mean $\pm$ 95% CI reported
Rollout budget	1000 MCTS simulations per action; search depth 50
Discount factor	$\gamma = 0.99$ for all planners
Exploration constant	$c = \sqrt{2}(R_{\max} - R_{\min})$ for PUCT baseline
Temperature grid	$\tau \in (0, 1]$ grid search; best τ chosen per game (ENT-regularised baselines)
Our hyperparameters	Atoms $N \in \{10, 20, 30, \ldots, 100\}$; power $p \in \{1, 2, 4\}$; bonus $C \in \{4, 8, 16\}$

G.2 Atari Hyperparameters

We run CATS in Atari with 10 random seeds, where each seed uses 1000 samples to collect the average score. We found that only 1000 simulations were necessary due to the utilization of a pretrained neural network. We run CATS with different numbers of atoms $\{20, 40, 60, 80, 100\}$ and select the best results.

The temperature parameter τ for MENTS and TENTS is tuned from the set:

$$\{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}.$$

The selected parameters are shown in Table 5. The exploration constant ϵ for MENTS and TENTS is set to 0.01. For Power-UCT, we select the power mean $p = 2$.

Table 5: Selected hyperparameter τ (temperature) for MENTS and TENTS across Atari games.

Game	MENTS	TENTS
Phoenix	0.07	0.60
MsPacman	0.09	0.03
Alien	0.10	0.03
SpaceInvaders	0.02	0.06
BeamRider	0.02	0.03
Robotank	0.01	0.05
Seaquest	0.02	0.03
Asteroids	0.08	0.20
Enduro	0.02	0.10
Atlantis	0.08	0.03
Hero	0.40	0.03
Frostbite	0.01	0.02

G.3 Pre-trained Value Network

We follow exactly the DQN recipe [Mnih et al., 2015]:

- **Architecture:** 3 convolutional layers (32×8, 64×4, 64×3) followed by 2 fully-connected layers ($512 \to |A|$).
- **Training:** 10M ALE frames per game; ε-greedy exploration with $\varepsilon : 1 \to 0.1$ over 1M steps; replay buffer size 1M; RMSProp with learning rate 2×10^{-4}.
- **Saved heads:** After training, for any state s we store $Q(s, a)$ and $V(s) = \max_a Q(s, a)$.

G.4 Node Initialization in Search

Following Xiao et al. [2019], we initialize node logits as shown in Table 6.

Table 6: Initial logits used by each planner at node expansion.

Planner	$\boldsymbol{Q}_{\text{init}}(\boldsymbol{s}, \boldsymbol{a})$
MENTS / TENTS	$\dfrac{Q(s,a) - V(s)}{\tau}$
RENTS	$\log P_{\text{prior}}(a \mid s) + \dfrac{Q(s,a) - V(s)}{\tau}$, where $P_{\text{prior}} = \mathrm{softmax}(Q(s, \cdot))$
CATSO / PATSO	Same shifted logits (with τ tuned as above)

Table 7: Sensitivity to the power-mean exponent p on SYNTHETICTREE with $(k, d) = (8, 3)$ and $n_{\text{sims}} = 1000$. Entries show mean root absolute error $\pm$ 95% CI over 10 seeds.

$\boldsymbol{p}$	**CATSO**	**PATSO**
1.0	0.230 ± 0.009	0.221 ± 0.015
2.0	0.229 ± 0.009	0.219 ± 0.015
4.0	0.227 ± 0.009	0.215 ± 0.015
8.0	0.225 ± 0.009	0.208 ± 0.015
∞	$\mathbf{0.189 \pm 0.012}$	$\mathbf{0.155 \pm 0.017}$

H Additional SYNTHETICTREE Ablations

This appendix reports the full SYNTHETICTREE ablation suite omitted from the main paper for space.

H.1 Hyperparameter Sensitivity

We sweep the power-mean exponent p and other key hyperparameters (C, N, K) on the tree configuration $(k, d) = (8, 3)$ with $n_{\text{sims}} = 1000$.

H.2 Component Ablations Across Tree Shapes

We summarize all six (k, d) pairs at $n_{\text{sims}} = 1000$ by comparing, for each tree, the best distributional max-backup variant (CATSO or PATSO, $p = \infty$) with the best non-distributional baseline (ScalarTSOpt or Power-UCT).

H.3 Effect of Stochasticity

On a fixed tree $(k, d) = (8, 3)$ we sweep reward and transition noise over four regimes and report root error at $n_{\text{sims}} = 1000$.

H.4 Runtime Details

We report per-simulation runtime comparisons and reproduce the exact timings using `runtime_benchmark.py`.

I Advantages Over Existing Methods

We clarify the conceptual and technical differences from prior work:

1. **Non-stationary bandit view at each node + distributional value backups.** We explicitly model each MCTS node as a non-stationary bandit (arm means drift because action selection changes which children are explored). We prove concentration *and* convergence of the *distributional* value estimator under this non-stationarity, culminating in non-asymptotic simple regret $\mathcal{O}(n^{-1/2})$ for both `CATSO` and `PATSO`. Prior TS+optimism MCTS papers either (a) analyze only expectations (no distributional backups), (b) do not handle the explicit non-stationarity induced by MCTS selection, or (c) give asymptotic/loose guarantees.

2. **Power-mean backup with analysis.** To our knowledge, our work is the first to provide a complete finite-sample analysis of MCTS with a *power-mean* backup [Dam et al., 2019] under TS+optimism and to prove concentration at *every internal node* (Theorems 3–4).

3. **No discretization error for `PATSO` + explicit bound for `CATSO`.** We show `PATSO` incurs zero representation error (it tracks the empirical distribution exactly), and for `CATSO` we give a clean

Table 8: Sensitivity to remaining hyperparameters on SYNTHETICTREE with $(k, d) = (8, 3)$ and $n_{\text{sims}} = 1000$. Entries show mean root absolute error $\pm$ 95% CI over 10 seeds.

Hyperparameter	Value	Error
C (PATSO)	0.5	0.210 ± 0.019
	1.0	0.210 ± 0.016
	2.0	0.222 ± 0.012
	4.0	0.227 ± 0.017
N (CATSO)	50	0.231 ± 0.008
	100	0.230 ± 0.009
	200	0.231 ± 0.007
	400	0.230 ± 0.010
K (PATSO)	50	0.221 ± 0.016
	100	0.216 ± 0.015
	200	0.221 ± 0.015
	400	0.221 ± 0.015

Table 9: Summary of component ablations on SYNTHETICTREE at $n_{\text{sims}} = 1000$. For each (k, d) we compare the best distributional TS variant with max backup (CATSO or PATSO, $p = \infty$) against the best non-distributional baseline (ScalarTSOpt or Power-UCT). Entries show mean root absolute error $\pm$ 95% CI over 10 seeds, plus relative gain of the distributional variant.

k	d	Dist. (max backup)	Best scalar baseline	Rel. gain (%)
14	3	PATSO (max), 0.139 ± 0.022	Power-UCT, 0.244 ± 0.012	42.9
16	1	PATSO (max), 0.044 ± 0.021	Power-UCT, 0.113 ± 0.016	61.5
16	3	PATSO (max), 0.135 ± 0.024	ScalarTSOpt, 0.253 ± 0.013	46.7
16	4	PATSO (max), 0.143 ± 0.023	Power-UCT, 0.258 ± 0.011	44.6
200	1	CATSO (max), 0.203 ± 0.017	Power-UCT, 0.203 ± 0.014	0.0
200	2	CATSO (max), 0.197 ± 0.034	Power-UCT, 0.252 ± 0.010	21.8

discretization error bound:

$$\mathcal{O}\left(\frac{R_{\max}}{N(1-\gamma)^2}\right),$$

and explain how to set N to match a target accuracy.

4. **WDRO link and sample complexity.** We connect our distributional MCTS to Wasserstein Distributionally Robust Optimization and derive a (loose but novel) sample complexity for (ε, δ)-robust policies—this angle is absent in the cited works.

J Limitations

- **Computational Demands:** The `CATSO` distributional Monte Carlo Tree Search (MCTS) faces challenges in managing computational demands while maintaining and updating probability distributions, leading to slightly increased complexity.
- **Fixed Precision:** The `PATSO` set of particles can increase in size if the observed values are different. We prevent this in the implementation by fixing the float precision.
- **Number of Atoms:** Our approach's performance is slightly influenced by hyperparameters, with the number of atoms being a critical factor. Suboptimal choices may affect performance.

K Details for Section 5.5: Memory Capping and W_1 Control

K.1 Merge-on-Insert (Precise Algorithm)

We keep a sorted container of pairs $\{(Q_i, \alpha_i)\}_{i=1}^{M}$ with $M \leq K$. Upon receiving a new sample z:

1. **Match:** If $\min_i |z - Q_i| \leq \varepsilon_{\text{tol}}$, let $i^\star = \arg\min_i |z - Q_i|$ and set $\alpha_{i^\star} \leftarrow \alpha_{i^\star} + 1$.

Table 10: Effect of stochasticity on SYNTHETICTREE with $(k, d) = (8, 3)$ at $n_{\text{sims}} = 1000$. We vary leaf reward noise σ and transition noise; entries show mean root absolute error $\pm$ 95% CI over 10 seeds.

Noise regime	CATSO	PATSO	ScalarTSOpt	Power-UCT
Deterministic	0.492 ± 0.007	0.447 ± 0.012	0.448 ± 0.012	0.359 ± 0.016
Low noise	0.359 ± 0.004	0.330 ± 0.005	0.344 ± 0.007	0.304 ± 0.007
Medium noise	0.226 ± 0.007	0.213 ± 0.009	0.221 ± 0.008	0.205 ± 0.013
High noise	0.078 ± 0.014	0.071 ± 0.018	0.073 ± 0.018	0.070 ± 0.017

2. **Insert:** Else if $M < K$, insert $(z, 1)$ and keep the list sorted.
3. **Merge:** Else (overflow): find j with smallest gap $|Q_{j+1} - Q_j|$, replace the pair by

$$\left(Q^{\text{new}}, \alpha^{\text{new}}\right) = \left(\frac{\alpha_j Q_j + \alpha_{j+1} Q_{j+1}}{\alpha_j + \alpha_{j+1}},\ \alpha_j + \alpha_{j+1}\right),$$

then insert $(z, 1)$ in sorted order.

K.2 Bound on W_1 Error Under Capping

Let $\widehat{P}$ be the (uncapped) empirical distribution and $\widetilde{P}$ the capped distribution after applying the procedure above. With nearest-neighbor merges:

$$W_1(\widehat{P}, \widetilde{P}) \ \leq \ \sum_{\text{merges}} (\alpha_j + \alpha_{j+1}) \cdot |Q_{j+1} - Q_j| \ = \ O\Big(\frac{\texttt{range}}{K}\Big).$$

Since the Bellman operator is $1/(1-\gamma)$-Lipschitz in W_1, the root-value error contributed by capping is $O\big(\texttt{range}/(K(1-\gamma))\big)$ and the simple-regret additive term is $O\big(\texttt{range}/(K(1-\gamma)^2)\big)$, which is dominated by the proven $O(n^{-1/2})$ term for moderate K.

L Additional Oracle-MDP Results

This appendix reports results for the full stochastic MDP suite and the full baseline set. In particular, we include *Gaussian-friendly controls* (e.g., additive-cost shortest-path) where Gaussian-return posteriors (DNG) are expected to be strong.

L.1 Evaluation Protocol

Across all environments, we evaluate each method by running MCTS from a fixed start state s_0 for a range of simulation budgets $n \in \{1, \ldots, N_{\max}\}$ and recording the recommended root action $\hat{a}_n$ (curves are plotted versus n on a log-x axis).

For oracle-evaluable tabular MDPs, we compute $V^\star(s_0)$ and $Q^\star(s_0, \cdot)$ via exact dynamic programming and report:

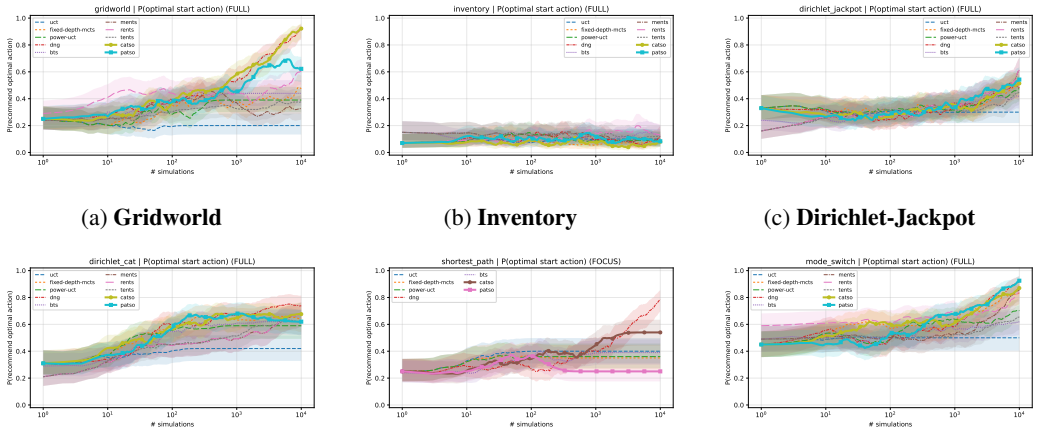
- Root simple regret: $\mathrm{SR}(n)$
- Probability of optimal action: $P_{\text{opt}}(n) = \mathbb{P}[\hat{a}_n \in \arg\max_a Q^\star(s_0, a)]$
- Root value error: $\mathrm{VE}(n) = |\hat{V}_n(s_0) - V^\star(s_0)|$

For SYNTHETICTREE, we report root absolute value error at a fixed budget (and sweeps over noise/hyperparameters). For Atari, we report mean episode return using 1000 simulations per decision step over multiple episodes and seeds.

Unless stated otherwise, results are averaged over independent random seeds (with shaded uncertainty bands/CI), and any smoothing is applied for visualization only.

L.2 Additional Environments

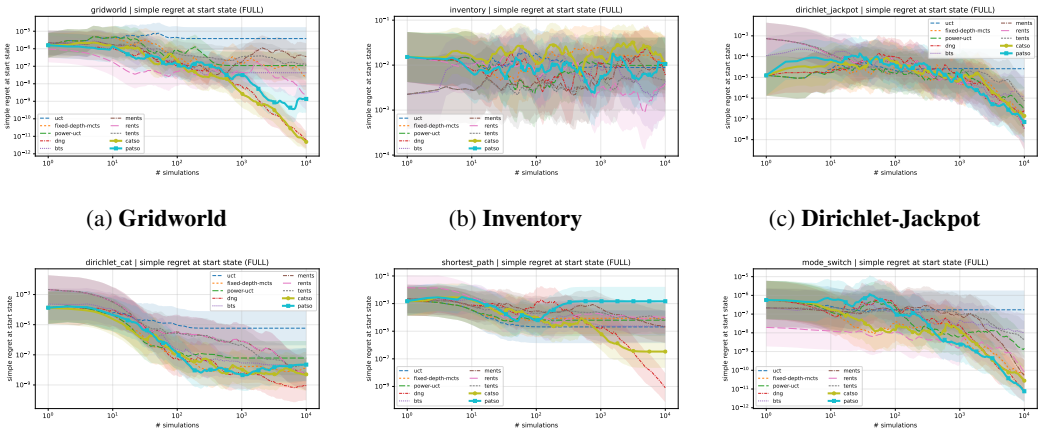
Beyond the two highlighted benchmarks, we evaluate on several oracle-evaluable stochastic MDPs that stress different sources of uncertainty:



(a) **Gridworld** (b) **Inventory** (c) **Dirichlet-Jackpot**

(d) **Dirichlet-Cat** (Gaussian-friendly) (e) **Shortest-Path** (Gaussian-friendly) (f) **Mode-Switch** @ larger budget

Figure 3: $P_{\text{opt}}(n)$ curves for additional environments (full baseline set).



(a) **Gridworld** (b) **Inventory** (c) **Dirichlet-Jackpot**

(d) **Dirichlet-Cat** (Gaussian-friendly) (e) **Shortest-Path** (Gaussian-friendly) (f) **Mode-Switch** @ larger budget

Figure 4: Root simple regret curves for additional environments (full baseline set).

- **Stochastic Gridworld:** A small navigation task with slip/noisy transitions (and optional traps/terminal rewards), designed to test robustness under transition stochasticity and long-horizon credit assignment.
- **Inventory Control with Mixture Demand:** Models ordering under uncertain demand, where per-step demand is drawn from a mixture distribution (capturing bursty or regime-dependent demand) and rewards include sales revenue, ordering costs, holding costs, and stockout penalties.
- **Random Dirichlet MDPs:** Transition kernels sampled from Dirichlet priors with terminal rewards that are either (i) categorical or (ii) "jackpot"-style with rare large payoffs, providing a controlled way to vary stochasticity, multimodality, and tail risk.

Complete MDP definitions and all parameter settings (state/action sizes, horizons, noise levels, mixture weights, and reward scales) are reported in the appendix.

L.3 Results: Probability of Optimal Action

L.4 Results: Simple Regret

L.5 When Gaussian-Return Baselines Excel

In environments where the cumulative return is close to Gaussian (e.g., additive costs over long horizons), Gaussian-return posterior methods such as DNG can be particularly strong. This behavior is expected (CLT-like averaging) and complements the main-text results, which target discrete, rare-event, and multi-modal return regimes. Figures 3 and 4 include such Gaussian-friendly controls (e.g., SHORTEST-PATH and DIRICHLET-CAT).